

UMEÅ UNIVERSITY

AID AND HEALTH
IN A CONTESTED BALANCE

Adopting effective development cooperation and improving health system governance in fragile and conflict-affected states: lessons from Sudan

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Umeå 2026

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Dissertation for PhD
ISBN: 978-91-8070-978-1 (print)
ISBN: 978-91-8070-979-8 (pdf)
ISSN: 0346-6612
Umeå University Medical Dissertations, New Series No 2416
Cover photo by Abdelmoneim Sayed / AFP
Electronic version available at: http://umu.diva-portal.org/
Printed by: Scandinavian Print Group, Hägersten
Umeå, Sweden 2026

To my fellow Sudanese, and to all who continue to suffer under conflict and tyranny, may your resilience light the path to peace

كان ياما كان في مرِّ الزمان
كان سلطانٌ وشرطيٌ وكاهن ومغنٍ
هبّت الثورةُ والسلطانُ فرّ
رجع الكاهنُ للمعبدِ يبكي
عمل الشرطيُ فلاحاً وكر
يطلق الماءَ على صدرِ الجداول
يضرب الأرضَ على صوت الكمان
وهو يهتزُّ ويلتذُ معَ خبطِ المعاول
كلّنا للأرضِ والأرضُ لنا
ولنا متسع حتى السماء
فإلام،
ولِم يغمر الأوطان طوفان الدماء؟
يا أخا الأرض
لِمَنْ هذه الحربُ؟
ومَنْ ذا الذي يشقى
ومَنْ يعشق الأوطانَ أطلال دمارٍ ودمن!؟
إنه البومُ
الذي يعجبه ليل الخراب
إنه الشؤم
الذي ينجبه ويلُ الحراب
هل أتي اليوم الذي تعقبه خيلُ الصحاب
فيحطُّ الغيمُ
في جوف أراضينا ويرتاح التراب

هل أتى حينٌ من الحلم
على الإنسانِ
يحيا مطمئناً للزمنْ؟!
تضع الحربُ به أوزارَها
كرنفال السلْمِ يجتاح مدى العالم
إبداعاً وقمحا
والعصافيرُ السلامية
تبني دارَها حباً وصدحا
يا ورودَ الدرب يا أحجارَها ليلاً وصبحا
يا خطى العمّال يا أيّارَها خبّاً وكدحا
لاتّحاد الساعدِ الكوني مرحى
لاخضرارات الفنْ
لانغراز الإصبع الثوري
في عين الدمار
وارتفاع العلبة البنبانَ كوباً من لبنْ
لتجار الحربِ
تبّاً..
للخوازيق
وتباً للبوار
للسرابيات في تيه الفِتن
للذي يُبحر غرباً عن أماني الشعب
تبّاً للدوار
للخطابياتِ من دون فِطن

لانطلاق الريح في غير لقاحْ
دون خيرٍ وصلاحْ
للتي تمطرُ موتاً وجراح
والتي تمرح من دون رسَنْ
في مراعي الشيبِ في حقلِ الشباب
للذي يملأ كفيه بتبرٍ وترابْ
يكشح التبرَ على حظوته
والترابيات في عين الوطن
أدعياءُ الحرب كالحرب تماما
أولياءُ السلم لا خوفٌ
عليهم لا حزَنْ
وإذا مرّوا بهم قالوا سلاما
وإذا غُمَّ بهم نادوا وطنْ
هذه الحربُ وثنْ
وأنا أؤمن
بالحرب التي بين أحشائي وتجويع الفجارْ..
الطواغيت... إذن
فلتكن يا عام عام الإنصهار
بالذي أدعوه قلبي

فلتكن يا عام عام الإنصهار
بالحبيبةِ، بالقريبةِ للوطن
ولتكن يا شعب ضد الإنكسار
وليكن ما قد دفعنا من دماء لانبلاج الفجرِ
محراً أو ثمن
نحن أقسمنا
على صدر عموم الشهداء الفقراء
أن نسوّيه بروحٍ وبدنْ
ونقوّيه على سوق النهار
ونضوّيه وطنْ
فلتكن يا عام عام الانتصار
للجماهير على بؤس الزمن
قد كفاها ما دهاها من محار
من عذابٍ ومحنْ
فلتكن يا بحر بحراً، للبحارْ
التي تنضحُ سلماً دون مَن
عاشت الأرض لنا لافتةً
ولأعداءِ العصافير كَفَنْ
هذه الحربُ وثنْ

لروح الشاعر العظيم محمد الحسن سالم حميد - من ديوانه ارضاً سلاح
ولكل محبي السلام، في السودان وكل مكان

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Abstract

Fragile and conflict-affected states (FCAS) face persistent challenges that undermine their ability to plan, finance and deliver essential services. Political instability, recurrent emergencies and fragmented external assistance (aid) weaken national institutions and constrain progress toward development targets like universal health coverage (UHC). Effective Development Cooperation (EDC) principles—ownership, alignment, harmonization, managing for results and mutual accountability—were introduced globally to address shortcomings of aid, yet evidence on their functioning and effects in FCAS remains limited. This thesis examined how these principles operate in practice, how they interact with health system governance, and whether they contribute to progress on health goals. Sudan, a long-standing FCAS characterized by shifting political regimes, contested legitimacy, and a highly aid-dependent health sector, served as the primary empirical case.

A mixed-methods research design was applied across four sub-studies. Three qualitative sub-studies explored the evolution of health sector partnerships in Sudan, drawing on semi-structured interviews conducted in 2015 and 2021/22 with government officials, donors, development banks, UN agencies, international NGOs and civil society organizations. These were complemented with document review and observations of coordination meetings. The fourth sub-study applied a quantitative comparative analysis across 17 aid-recipient FCAS to examine whether specific EDC principles were associated with advancing UHC.

The findings revealed shifting patterns of partnership behavior shaped profoundly by political context. In Sudan, periods of authoritarian rule constrained alignment and partners' willingness to use national systems, while the post-2019 transitional government created a window of opportunity for advancing collaboration due to increased transparency and greater alignment with national priorities. However, unfortunate political reversals, notably the 2021 coup, rapidly dismantled progress, triggering withdrawal of donor support and re-crumbling of coordination. While earlier the civil society organizations were marginalized by donors and other global partners, and manipulated by the authoritarian regime, they were soon aspired – by the global partners – to assume a prominent implementation role when relation with the state was cut after the coup.

Adopting the Humanitarian–Development Nexus (HDN) was sought as a policy solution to ensure development in the midst of recurrent emergencies and humanitarians' presence, but it had its own challenges. While widely praised, its operationalization was hindered by conceptual ambiguity, tensions between humanitarian and development actors, and unresolved debates about system utilization. A persistent dilemma materialized between strengthening national systems and using them. Humanitarian agencies often maintained parallel mechanisms for procurement, financial management and information systems, citing weak national capacity and the exigencies of crisis response. Development partners and government increasingly advocated a "use to strengthen" approach, emphasizing that durable improvements require channelling assistance through national systems. Over time, some partners, including several UN agencies and international NGOs, shifted toward more active system-strengthening and selective utilization, supported by joint technical assistance and piloting of harmonized approaches.

The cross-country analysis of FCAS demonstrated two principles—alignment with national priorities and joint technical assistance—were individually sufficient to attain UHC in several FCAS. The findings illustrates that multiple pathways can lead to improved outcomes, and that targeted, coordinated support to core governance functions can yield development even amid fragility.

The thesis concludes that EDC has the potential to influence governance and accelerate health system gains in FCAS, yet its effectiveness is highly contingent on political dynamics and functioning coordination platforms. Progress depends on inclusive national ownership, coherent and flexible technical assistance, and gradual yet deliberate movement toward use of national systems. Empowering civil society, maintaining coordination fora, and strengthening monitoring and the accountability mechanisms are essential to sustain gains, particularly in FCAS volatile environments. Findings highlight that although EDC cannot resolve structural fragility, it can create reinforcing pathways that transform fragmented aid into more coherent systems support when global and national actors engage with political realities while upholding principled cooperation.

Keywords: effective development cooperation; fragile and conflict-affected states; health system strengthening; governance; universal health coverage; humanitarian–development nexus.

الملخص باللغة العربية

تواجه الدول الهشّة والمتأثرة بالنزاعات تحديات مستمرة تقوّض قدرتها على التخطيط والتمويل وتقديم الخدمات الأساسية. فعدم الاستقرار السياسي، وتكرار الأزمات والطوارئ، وتشتت المساعدات الخارجية، يُضعف المؤسسات الوطنية ويحدّ من التقدم نحو أهداف التنمية مثل التغطية الصحية الشاملة. طورت مبادئ فعّالية التعاون الإنمائي - الملكية الوطنية، المواءمة، التنسيق، الإدارة الموجهة نحو النتائج، والمساءلة المتبادلة - لمعالجة أوجه القصور في المساعدات والتعاون الإنمائي، إلا أن الأدلة المتعلقة بمدى عمل هذه المبادئ وتأثيرها في الدول الهشّة ما تزال محدودة. تبحث هذه الأطروحة كيفية تطبيق هذه المبادئ عمليًا وكيفية تفاعلها مع حوكمة النظم الصحية، وما إذا كانت تسهم في إحراز تقدم نحو الأهداف الصحية.

شكّل السودان الحالة التجريبية الأساسية للدراسة، وهو دولة هشّة تتسم بتغير الأنظمة السياسية والحروب وتنازع الشرعية واعتماد كبير على المساعدات في الرعاية الصحية وبرامج الصحة العامة الاساسية.

استخدم البحث تصميمًا يعتمد الأسلوب المختلط عبر أربع دراسات. استكشفت ثلاث دراسات نوعية تطور الشراكات في القطاع الصحي في السودان، واستندت إلى مقابلات شبه مهيكلة أُجريت عامي 2015 و22/2021 مع مسؤولي الحكومة والجهات المانحة والبنوك التنموية ووكالات الأمم المتحدة والمنظمات الدولية غير الحكومية ومنظمات المجتمع المدني. وقد استُكملت هذه البيانات بمراجعة الوثائق وملاحظات من حضور اجتماعات منصات التنسيق. أما الدراسة الفرعية الرابعة فقد استخدمت تحليلًا كميًا مقارنًا عبر 17 دولة هشّة متلقية للمساعدات لفحص ما إذا كانت مبادئ فعّالية التعاون الإنمائي مرتبطة بتحقيق التغطية الصحية الشاملة.

تكشف النتائج عن أنماط متغيرة في سلوك الشراكات تتأثر بصورة عميقة بالسياق السياسي. ففي السودان قيدت فترات الحكم السلطوي إمكانية المواءمة واستعداد الشركاء لاستخدام الأنظمة الوطنية، بينما أوجدت الحكومة الانتقالية بعد عام 2019 نافذة فرصة لتعزيز التعاون نتيجة زيادة الشفافية مع ارتفاع رغبة الشركاء الدوليين في المواءمة مع الأولويات الوطنية. إلا أن الانتكاسات السياسية اللاحقة لانقلاب 2021 سرعان ما قوضت ما تحقق من تقدم، وانعكس ذلك على إيقاف برامج وانسحاب مانحين وانهيار قنوات التنسيق. وفي حين كان المجتمع المدني يُهمّش سابقًا من قبل الجهات المانحة وشركاء عالميين آخرين، بل ويُستغل من قبل النظام الحاكم، أصبح (بعد قطع العلاقات مع الدولة عقب الانقلاب) محط اهتمام الشركاء العالميين الذين سعوا إلى منحه دورًا بارزًا في التنفيذ كبديل عند ايقاف التعامل مع نظام الانقلاب.

وقد جرى الدفع نحو تبني نهج "الربط بين العمل الإنساني والتنمية" كحل سياسي لضمان استمرار التنمية في سياق الأزمات المتكررة والحضور الكثيف للفاعلين الإنسانيين، إلا أن تطبيقه واجه تحديات خاصة. فعلى الرغم من الإشادة الواسعة بهذا النهج فقد أعاق عدم وجود توافق مفاهيمي حوله، والتوترات بين الفاعلين الإنسانيين والإنمائيين، والجدل غير المحسوم حول كيفية استخدام الأنظمة الوطنية، اعاق عملية تفعيله. كذلك برزت معضلة الجدال بين تقوية الأنظمة الوطنية وبين استخدامها. فقد حافظت الجهات الإنسانية في كثير من الأحيان على آليات موازية للمشتريات والإدارة المالية وأنظمة المعلومات، مستندةً إلى ضعف القدرات الوطنية وضرورات الاستجابة للأزمات العاجلة.. بالجهة المقابلة دعا الشركاء الإنمائيون والحكومة خاصة بشكل متزايد إلى منهج "الاستخدام من أجل التقوية"، مشددين على أن التحسينات المستدامة تتطلب تمرير المساعدات عبر الأنظمة الوطنية. ومع مرور الوقت، تحوّل بعض الشركاء (بما في ذلك عدة وكالات أممية ومنظمات دولية غير حكومية) نحو دور أكثر فعالية في تقوية الأنظمة واستخدامها بشكل انتقائي مرتكزين على نجاح مساعدات فنية مشتركة وتجارب لنهج عمل منسّق.

أظهر التحليل المقارن عبر الدول أن مبدأي - المواءمة مع الأولويات الوطنية، والمساعدة الفنية المشتركة - كانا كافيين بمفردهما لتيسير تحقيق التغطية الصحية الشاملة في عدة دول هشّة. وتوضح النتائج أن مسارات متعددة يمكن أن تقود إلى تحسين النتائج، وأن الدعم المستهدف والمنسق لوظائف الحوكمة الأساسية يمكن أن يحقق التنمية حتى في ظل الهشاشة.

تخلص الأطروحة إلى أن فعّالية التعاون الإنمائي يمكن أن تؤثر على الحوكمة وتعجل من مكاسب الأنظمة الصحية في الدول الهشّة، إلا أن فاعليتها تعتمد بدرجة كبيرة على الديناميكيات السياسية وعلى عمل منصات التنسيق. ويتوقف التقدم على ملكية وطنية شاملة، ومساعدات فنية متماسكة ومرنة، مع حركة تدريجية ومدروسة نحو استخدام الأنظمة الوطنية. كما يعد تمكين المجتمع المدني، والحفاظ على منصات التنسيق، وتعزيز آليات المتابعة والمساءلة عناصر أساسية لاستدامة المكاسب، خصوصًا في البيئة المتقلبة للدول الهشّة. تبرز النتائج أنه على الرغم من أن مبادئ فعّالية التعاون الإنمائي لا يمكنها حل الهشاشة البنيوية لهذه الدول، إلا أنها قادرة على خلق مسارات معزّزة تحول المساعدات إلى دعم أكثر اتساقًا للأنظمة عندما ينخرط الفاعلون الوطنيون والدوليون في تعاون مبدئي لا يهمل واقع السياق السياسي.

الكلمات المفتاحية: فعالية التعاون الإنمائي، الدول الهشة والمتأثرة بالنزاعات، تقوية النظام الصحي، الحوكمة، التغطية الصحية الشاملة، الربط بين العمل الإنساني والتنمية.

Abbreviations

Abbreviation	Full Form
ALNAP	Active Learning Network for Accountability and Performance in Humanitarian Action
BMGF	Bill & Melinda Gates Foundation
CEPI	Coalition for Epidemic Preparedness Innovations
CPA	Comprehensive Peace Agreement
CSO / CSOs	Civil Society Organization(s)
COVID-19	Coronavirus Disease 2019
DAC	Development Assistance Committee of the Organization for Economic Co-operation and Development
DAH	Development Assistance for Health
DHIS2	District Health Information System 2
DRC	Democratic Republic of Congo
EDC	Effective Development Cooperation
EMRO	Eastern Mediterranean Regional Office for the World Health Organization
EU	European Union
FCAS	Fragile and Conflict-Affected States / Situations
FMoH	Federal Ministry of Health (Sudan)
GAVI	Global Alliance for Vaccines and Immunization
GDP	Gross Domestic Product
GHIs	Global Health Initiatives
GFF	Global Financing Facility
GNI	Gross National Income
GPEDC	Global Partnership for Effective Development Co-operation

HACT	Harmonized Approach to Cash Transfers
HDN	Humanitarian–Development Nexus
HDPNx	Humanitarian–Development–Peace Nexus
HIS	Health Information System
HSPF	Health Sector Partners Forum
ICC	International Criminal Court
ICT	Information & Communication Technology
IDA	International Development Association
IHP+	International Health Partnership
IMF	International Monetary Fund
INGO(s)	International Non-Governmental Organization(s)
IRC	International Rescue Committee
JAR	Joint Annual Review
LIC	Low-Income Country
LMIC	Lower-Middle-Income Country
MDGs	Millennium Development Goals
M&E	Monitoring and Evaluation
MoH	Ministry of Health
MSF	Médecins Sans Frontières
NGO(s)	Non-Governmental Organization(s)
NMSF	National Medical Supplies Fund (Sudan)
OCHA	United Nations Office for the Coordination of Humanitarian Affairs
ODA	Official Development Assistance
OECD	Organization for Economic Co-operation and Development
PEPFAR	United States of America President's Emergency Plan for AIDS Relief

PFM	Public Financial Management
PMU(s)	Project Management Unit(s)
PPP(s)	Public-Private Partnership(s)
QCA	Qualitative Comparative Analysis
RCov	Raw Coverage
UCov	Unique Coverage
RSF	Rapid Support Forces
SAF	Sudan Armed Forces
SDG(s)	Sustainable Development Goal(s)
SDG3 GAP	Global Action Plan for SDG 3
TA	Technical Assistance
U5MR	Under-Five Mortality Rate
U5MRC	Under-Five Mortality Rate Relative Change
UHC	Universal Health Coverage
UHC2030	UHC2030 Global Partnership
UN	United Nations
UNAIDS	United Nations Programme on HIV/AIDS
UNDP	United Nations Development Programme
UNFPA	United Nations Population Fund
UNICEF	United Nations Children's Fund
UN Women	United Nations Entity for Gender Equality
U.S.	United States of America
USAID	U.S. Agency for International Development
WFP	World Food Programme
WHO	World Health Organization

Glossary

Aid	To provide help and/or assistance of a practical nature (Oxford English Dictionary, n.d.). In terms of the Official Development Assistance (ODA) "those flows to countries and territories and to multilateral institutions which are provided by official agencies, including state and local governments, or by their executive agencies; and each transaction of which is administered with the promotion of the economic development and welfare of developing countries as its main objective; and is concessional in character and conveys a grant element of at least 25%" (Organization for Economic Co-operation and Development [OECD], 1969). In terms of Humanitarian aid, it is a form of assistance designed to save lives and alleviate suffering during and after crises, such as floods, famine or conflicts (International Rescue Committee, 2025).
Development Assistance for Health (DAH)	The financial and in-kind resources provided by external donors, such as governments, multilateral organisations and private foundations, to low- and middle-income countries to support health programs and improve outcomes (Xie et al., 2025)
Effective Development Cooperation (EDC)	A framework guiding equitable, transparent and results-oriented development partnerships, formally used instead of aid effectiveness starting at Busan high-level forum (OECD, 2011)
Fragile and Conflict-Affected States (FCAS)	Countries and territories that fall within one or combination of the characteristics of ongoing conflict, weak or contested governance institutions, and limited capacity to deliver essential services (World Bank, 2025) .
Health System Governance	The processes, structures and institutions and authority that are in place to oversee and manage the health system (World Health Organization, 2011)
Humanitarian–Development–Peace Nexus (HDPN)	An approach linking humanitarian relief with long-term development and peacebuilding efforts (World Health Organization., Regional Office for the Eastern Mediterranean [WHO EMRO], 2021)

List of papers

1. **Aweesha, H.**, Hurtig, A., Pulkki-Brännström, A., & San Sebastian, M. 2024. Sudan's health sector partnership: From confined progression to openness and hope to uncertain demise. *Development Policy Review*, *42*(2), Article e12757. https://doi.org/10.1111/dpr.12757

2. **Aweesha, H.**, Hurtig, A., Pulkki-Brännström, A, & San Sebastian, M. 2025. Humanitarian–development nexus approach to health systems strengthening in Sudan—a policy analysis. *Frontiers in Public Health*, *13*, Article 1579825. https://doi.org/10.3389/fpubh.2025.1579825

3. **Aweesha, H.**, Pulkki-Brännström, A, Hurtig, A., & San Sebastian, M. Attitudes and practices of Sudan's health sector partners towards strengthening and utilising the national health system components (manuscript submitted, under review)

4. **Aweesha, H.**, Pulkki-Brännström, A, Hurtig, A., & San Sebastian, M. Pathways to Universal Health Coverage: A Qualitative Comparative Analysis of Effective Development Cooperation Principles in Fragile and Conflict-Affected States (manuscript submitted, under review)

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Prologue: life, career and politics entangled

I was born in Sudan in 1989, in the shadow of another failed democratic endeavour and the rise of an authoritarian military-Islamist regime. This regime imposed a rigid islamo-arab identity on a country of vast diversity —over 300 tribes, more than 100 languages, and multiple religions— while restricting civic space and reshaping social norms. Globally, the Cold War had just ended, and the new global order single pole brought harsh economic sanctions onto Sudan. Against this backdrop of political repression and economic decline, my childhood began from a place of relative privilege—until life took turns that would reshape everything.

Sanctions, corruption, and the long war in the South deepened the national crisis. Yet my parents provided me a life path many Sudanese could not dream of. Both were first-generation university graduates, my father a professor of Sharia and law, and my mother a holder of dual academic qualifications. Their political involvement within the Islamist movement profoundly shaped my upbringing, however conflicted with many of my later life choices. I attended a newly piloted "Quranic school," where memorizing and reciting the Quran dominated our days, with some explanation and understanding (not so much of the later I would argue). By sixth grade I had completed memorization and was celebrated in ways no other academic achievement of mine ever matched. This opened doors to elite secondary schools I might not have otherwise accessed despite my qualifying academic grades.

In secondary school, while religious teaching time diminished, my questions grew. I doubted rigid explanations of set values and religious judgements, quietly as caution to share doubts was critical in a judgmental environment. Alternatively, I turned to theology and philosophy. Whenever I asked my father for clarity, he rarely answered directly; instead, he pointed me to books in his large library. Only later did I recognize this as deliberate—a way to help me develop independent thought rather than inherit beliefs unquestioned.

As university approached, I had no strong preference and gravitated between the traditional Sudanese preferences, medicine or engineering. Knowing my parents hoped I choose medicine, I briefly leaned towards engineering, partly out of rebellion. Ultimately, I made a pragmatic decision when my highest grades that year were unexpectedly in biology, I accepted it as a sign and pursued medical school.

Joining the University of Khartoum marked a turning point. Known not only for academic excellence but also for political vibrancy, it was the last major university whose student union had not been taken over by the Islamist movement. I watched opposition leaders debate fiercely, then be chased and beaten by regime forces—many of them later becoming figures in the post-2019 transitional government.

The year 2005, when I started my university journey, marked the beginning of a transformative period in Sudan's history as the Comprehensive Peace Agreement (CPA) ended Africa's longest civil war. This led to opening of freedom spaces and improving global engagement as Sudan's oil reached global markets. Internet access expanded, giving people, including me, space to explore ideas beyond the regime's-imposed worldview. Navigating these openings opposite to the overall restrictions reshaped my political awareness and sense of civic responsibility.

Academically, I struggled to connect with biomedical science, yet I found meaning in extra-curricular activities of the medical students' association especially the public health committee, which I later led nationally and at the African regional level. Public health opened the world to me: international conferences, new people, new ways of thinking. It allowed me to bypass the regimes' isolation bubble. But politically, it placed me in a "no-man's land": opposition students distrusted my family background, and Islamist peers distrusted my questioning of their ideas. Still, up to this stage I took no political side.

By my graduation time, Sudan was descending once again—South Sudan's secession drained resources, global support faded, and crises multiplied. Motivated by what I witnessed in hospitals, preventable suffering and a collapsing system, I solidified my choice of public health over clinical medicine. My parents opposed this "madness," but I persisted.

I joined Sudan Ministry of Health as planning and monitoring officer within the directorate of primary health care. As part of the team finalizing Sudan's first comprehensive health map, I spotted stark inequalities beyond the urban capital boundaries. As I later moved to the Directorate of Planning and International Health, I did not question the combination of these functions as I have already understood the degree of dependence on global partners assistance for core health programs like immunization, malaria and maternal health.

After a few years, I realized I needed further knowledge and skills. I applied to a master's program in Umeå, a place I couldn't even locate on a map until planning my trip. How little did I know, I will meet snow! Moving from Sudan's searing heat to this Norrland magical city was a life changing experience. For this time, I truly enjoyed studying. When thesis writing emanated, I went to a teacher who always challenges our thoughts. Professor Miguel contested me to look deeper into the systemic issues of my homeland. Our discussions sparked memories of the centrality of aid, and its operational challenges. That became my studies topic of focus. Upon returning to Sudan that summer I contacted people I knew to explore the possibility of conducting this research. Everyone advised me to reach to Dr. Imad Kayuna, Director of International Health at Sudan Ministry of Health back then. He enthusiastically supported my fieldwork. Dr Imad later became my direct supervisor in post-master endeavours. The thesis earned an award and became the seed for my doctoral studies. As I finalized and shared the master thesis with Sudan ministry of health, I received a call from Dr Imad expressing interest in the findings and recommendations and proposing a meeting when I come back to Sudan.

Soon as I left Umea, thinking it is the end of my time in this precious city, I found that I would be asked to help design a comprehensive partnership coordination forum for the health sector. Throughout 2016 second half we built documents, negotiated, revised, and finally launched the Sudan Health Sector Partners Forum, which I coordinated for another year. Meanwhile I kept an eye on work on the local level. I invested all my remaining energy and passion on designing and supporting interventions on strengthening the decentralized health system at sub-national levels, for which I served as a coordinator for a specialised working group. During this tenure, I visited all Sudan 18 states, and I would proudly say that I had some capacity building or coordination role to play in all of them. Later, I joined UNICEF as a seconded technical support to the Ministry of Health focusing on enhancing PHC through better management and collaboration at local levels. By the end of 2018 I moved to WHO Sudan as a health systems officer, where I worked on same issues for five years.

At this time-point, the dazzling December revolution in Sudan took place, and with it, as political interest grew in me, I joined a revolution leading political party that align with my ideology. This caused significant friction with my parents, leading to heated debates, especially with my mother. I wasn't the only one supporting the revolution among my siblings though!

The revolution succeeded in removing the regime from power, ushering a period of immense hope where the sky was the limit for our aspirations. Up to this day I am part of this party and has been humbly serving at its health secretariat for the past couple of years. Regardless of the tragic changes in context, I continue to be proud of what we did back then.

As the revolution brought aspirations for advancements, I thought of ways to advance my career, knowledge and skills. A doctoral opening in Umeå appeared, and I applied eagerly, building on my earlier master thesis and the emerging developments following the SDGs era. I told my wife that perhaps finally I will show her that place I talk about a lot. I was accepted, again with Miguel, along the intelligent Anni-Maria with support from the insightful Anna-Karin. It was a sandwich program—ideal for continuing my work in Sudan. We arrived in Umeå for my registration visit, and one week later the world shut down due to COVID-19.

We returned to Sudan six months later, and by late 2021 I had completed most the interviews I needed at that stage. Everything looked promising. My country was recovering, my career progressing, and my doctoral work on track—until the morning of October 25th, when a military coup shattered the transition. Partners withdrew, collaborations collapsed, and some study questions became obsolete. Talk about mutual accountability how now when the partners cannot and do not want to talk to each other! I pivoted toward studying the humanitarian-development nexus and collected new interviews to reflect the altered landscape. I kept the hoped. Again, little I knew, the worst was yet to come.

In March 2023, I travelled alone to Umeå for what was hoped to be a short writing visit. Ramadan had begun, and I spent evenings watching Sudanese television shows to feel closer to home. On the night of Friday April 14th, I stayed up late and had a call with my wife and daughter. They were visiting family as we used to on Fridays and were convinced to stay. The night was peaceful, I felt, but it was the last of its kind. Next morning, I woke to news of war erupting in Sudan between the coup's two heads. Only after my family reached safety after few weeks that did I fully grasp the human weight of conflict—beyond anything I had read or imagined.

To this end I feel my life journey resembles what I found on Sudan health sector partnership journey: a story of progression despite constrains, followed by a period of hopeful openness, and then sudden tragic demise. Yet, like Sudan, the story is not finished.

Introduction

Fragile and conflict-affected States (FCAS) experience persistent instability, humanitarian crises, and fragmented aid environments that weaken state capacities and impede progress toward development goals. Yet, essential services, like health, within these settings rely heavily on external assistance, i.e. aid. The magnitude and multiplicity of partners that comes with aid present another challenge, with varying interests and power dynamics. These manifestations are well demonstrated in the health sector, with weakened health systems and array of health partners, hindering the progress toward main health targets like Universal Health Coverage (UHC). To address these challenges and assist FCAS to achieve their development targets, a set of principles are theorised to ensure Effective Development Cooperation (EDC). The EDC principles of ownership, alignment, harmonization, management for results and mutual accountability are globally promoted as partnership foundations. However, their practical application in volatile settings remains poorly understood, especially where institutions are fragile and political legitimacy is contested. Within this thesis, a deep dive into these issues is presented through the available evidence.

Sudan highly illustrates these complexities of fragility, volatile context and aid partnerships as will be seen at the background and the findings. Sudan health sector has long been characterized by dependence on external assistance, dominance of parallel humanitarian delivery systems, and sharp fluctuation in donor engagement driven by political transitions, including the 2019 revolution and shortly after the 2021 military coup. Meanwhile, Sudan has formally adopted cooperation frameworks such as the International Health Partnership (IHP+), the Sudan Health Compact, UHC2030, and the Humanitarian–Development-Peace Nexus (HDPNx). All these frameworks take the EDC principles at their core, offering a unique context in which to examine how such EDC commitments interact with unstable governance conditions and various partners' behaviours.

This thesis investigates how aid and health partnerships operate within the context of FCAS. Exploring how principles of EDC are practiced and how they shape and interact with the governance of the health system(s) in fragile settings is the focus, with Sudan being the central empirical case. The study further examines if and how these principles can advance UHC.

The thesis uses a mixed-methods design combining qualitative case study work based on Sudan with a cross-country comparative analysis. Three qualitative sub-studies draw mainly on interviews, supplemented by documents review and observations conducted in 2015 and 2021/22 to: (1) examine how health partnerships evolved throughout political periods, (2) how the Humanitarian-Development Nexus approach was introduced and operationalized, and (3) how partners supported and engaged with or bypassed the national system components of financial management, procurement, and health information. The fourth quantitative sub-study uses qualitative comparative analysis (QCA) across seventeen other FCAS to assess whether specific EDC principles—or any combinations of them—were associated with meaningful progress toward UHC.

Thesis outline

The thesis begins with a background chapter that situates the study within FCAS and their characteristics, the global debates on aid and development cooperation, health systems governance, and the humanitarian–development divide and nexus. It then culminates with a contextual overview of Sudan's political trajectory and health system. This is followed by a section outlining the study rationale, overarching aim, and specific objectives. Then I present the utilized theoretical framework positioning EDC principles as inputs that can strengthen governance when supported by appropriate contextual conditions such as political openness, partners coordination, emergencies and stability. The methods chapter describes the qualitative research conducted in Sudan and the cross-country QCA, detailing data collection, analytical approaches, and ethical considerations. The results chapters present the findings from all four sub-studies, examining how EDC principles were interpreted and operationalized in connection to how partners' behaviours shifted across political periods, how the HDN approach evolved, how national systems were strengthened, utilized or bypassed, and which combinations of principles were associated with improved UHC outcomes across FCAS. The discussion synthesizes these findings to highlight the centrality of political context, coordination fora, and aligned joint technical assistance in enabling EDC to improve cooperation towards development targets. The thesis concludes with recommendations for governments, donors, and global health partnerships on improving the practice and monitoring of EDC in fragile states and identifies critical avenues for future research.

Background

Fragile and conflict-affected states

The global landscape has witnessed a notable rise in the number of countries classified as Fragile and Conflict-Affected States (FCAS) in recent years (Martineau et al., 2017). Although definitions of fragility vary, common features of FCAS include the erosion of state legitimacy, loss of territorial control or monopoly over the use of force, and the inability to deliver essential public services (Fragile States Index, n.d.). These characteristics collectively undermine the capacity of states to maintain stability and meet the basic needs of their populations.

FCAS represent some of the utmost complex and challenging environments for achieving development goals (Bogale et al., 2024). FCAS mostly remain within this classification, moving between fragility status and conflict presence. Nearly three-quarters of the 39 FCAS classified by 2025 have remained classified as such for over a decade (Hill et al., 2025). FCAS settings are home to over one billion people, predominantly residing in low-income and lower-middle-income countries. The majority of these populations live below the poverty line and face high levels of displacement, with limited access to essential services. The demand for social services in FCAS often exceeds the capacity of the weakened state institutions. It is estimated that nearly half of global child deaths occur in these contexts, largely due to disruptions in preventive services and livelihoods (Martineau et al., 2017). Essential services systems like health in FCAS are further strained by recurrent emergencies, including infectious disease outbreaks, which compound the already high burden of injuries and illnesses (Bogale et al., 2024). These emergencies frequently overwhelm existing capacities and contribute to the destabilization and fragmentation of national systems (Khatri et al., 2023).

FCAS economies suffer severely limited fiscal space, which prevents them from spending adequately on essential public services, including education, health, and infrastructure (Hill et al., 2025). This leads to high dependence on external support (aid) in the provision of these services (Carment & Samy, 2023). Hence, FCAS witness proliferation of global actors and initiatives that are uncoordinated and mostly donor driven. Within these settings lies a further diverse composition of partners, with higher concentration of humanitarian actors compared to other settings.

The humanitarian actors have a high tendency to bypass national systems in favour of direct service delivery (Spicer et al., 2020). In their endeavour, humanitarian organizations and actors often prioritize short-term relief over long-term sustainability and system strengthening (Saez et al., 2021). Hence, despite all aid directed toward the services, evidence of progress on the development targets in FCAS remains limited (Leunig et al., 2024).

Aid: power politics or development partnerships?

Aid has long occupied a contested space in international relations and development studies and practice, oscillating mostly between being considered a strategic instrument of power politics or a mechanism for fostering genuine partnerships for development and relief (Gamso, 2025). Historically, donor countries have used aid to advance geopolitical interests, secure alliances, and exert influence over recipient states (Apodaca, 2017). Yet, after World War 2 and further later at the turn of the present millennium, geopolitical changes and economic pressures - discussed further below - necessitated shifts in aid theory, administration and collaboration measures at both donor and recipient countries (Addison & Scott, 2011; Lai, 2003; Mascarenhas, 2025).

Aid gained special political importance during the times of the Cold War (from World War 2 end in 1945 up to the Soviet Union Dissolution in 1991) as development assistance was considered an essential strategy to gain and sustain alliances (Lee, 2022). The Marshall Plan (1948), through which the United States of America (U.S.) provided $12 billion to rebuild World War 2 torn Europe, was not only an economic recovery initiative but also a strategic tool to counter Soviet influence and promote liberal values (McCourt & Mudge, 2023). The aid recipients in this case, as well in other cases, were driven to align with the donors' interest, hereby the U.S.-led international order, and the plan success unified Western Europe as U.S. allies against communism. Beyond Europe, the U.S. extended aid to neutral and non-aligned countries during the Cold War to sway their geopolitical orientation. Lee (2022) noted that between 1955 and 1970, the U.S. provided more unconditional aid to non-aligned states than to its formal allies, revealing attention to further political gains. Aid was often used to secure support in international forums, including voting in the United Nations (UN), allowing military presence, and maintaining access to strategic resources, not only by the U.S., but also by the Soviet Union and other power houses (Apodaca, 2017; Gamso, 2025; Wittkopf, 1973).

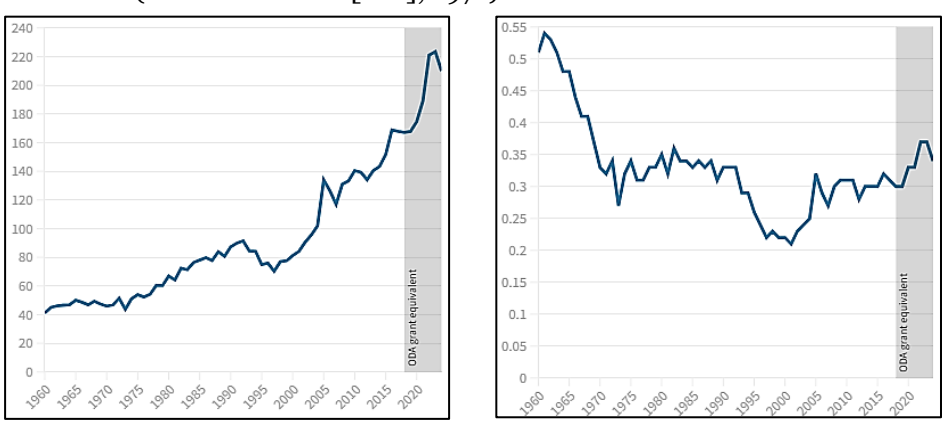
Under this view, the U.S. Agency for International Development (USAID) was founded in 1961 by President John F. Kennedy as a soft power tool (Maxwell, 2022). Aid soft power was also manifest in cultural diplomacy. Media programs and educational exchanges were designed to project American values and counter Soviet propaganda (Bu, 1999). Nye's concept of soft power—persuading others through attraction rather than coercion— (Nye, 2021) was embodied in these efforts, which complemented economic aid with ideological outreach. USAID is not the only example of such aid influence (Blair et al., 2022) but is probably the most striking, given its huge portfolio and amount of funds.

In recent years, foreign aid has continued to serve strategic interests. For instance, the U.S. used humanitarian assistance in countries like Sudan, Syria and Iran, not only to address crises, but also to maintain influence and prevent adversaries from filling power vacuums (Adrison, 2024; Ogunnowo & Chidozie, 2020). China's growing aid footprint, particularly in Africa and Southeast Asia, reflects a similar use of aid as a soft power tool, often tied to infrastructure projects (Blair et al., 2022).

On the other hand, as decolonization accelerated during the post-World War 2 era, international relations discourse shifted to address the needs of newly independent states (Birn et al., 2017a). The Marshall Plan success laid the groundwork for a common belief that aid could be a good catalyzer for development in these new emerging independent poorer countries (Mavrotas, 2009). Aid was embedded in modernization theory, which stipulated that underdeveloped countries could progress through stages of economic growth with external support (Lebo, 2021). Despite critiques, this theory continues to inform global policy, framing aid as a necessary—even if imperfect—tool for achieving development outcomes. This period witnessed the establishment of key international institutions like the United Nations, World Bank, and the International Monetary Fund (IMF), which all formalized aid as a development cooperation strategy towards achieving the global development agenda (Birn et al., 2017a). This development cooperation model has kept aid as part of global frameworks like the ended Millennium Development Goals (MDGs) and the current Sustainable Development Goals (SDGs), as the last goal within both was to enhance partnership for development, addressing the issue of aid directly (Goldin, 2016; Keeley, 2012).

Major critiques presided over this view of aid as a main development tool. For instance, aid dependency theory emerged stressing how the global economic structures perpetuated underdevelopment through unequal power relations, where aid often reinforced dependency and served mostly the interests of the donor countries (Goldin, 2016; Kabonga, 2017). Recipient countries, especially those who get aid for prolonged times fall into the trap of reliance on aid rather than focus on developing sustainable domestic systems for revenue generation, governance, and services. Governments in such contexts prioritize securing aid rather than implementing reforms or mobilizing domestic resources (like taxes), and answer mainly to donors, undermining their accountability to the citizens (Atitianti, 2025; Compaoré & Tagem, 2022; Knack, 2001). Further, foreign aid was scrutinized for being a tool for resource extraction by enabling donor states to secure critical minerals and rare earth elements, as well as oil and gas, while reinforcing post-colonial economic power relations (Akhigbodemhe & Azubuike, 2025; Ufimtseva et al., 2024).

Official Development Assistance (ODA) trends have fluctuated through various political and economic events since their recording by the Organization for Economic Co-operation and Development (OECD) from the 1960s. Figure 1 below depicts the overall trend in ODA since 1960 in U.S. Dollars, as well as ODA as percentage of Growth National Income (GNI) of donor countries, as per OECD (2025a). The UN International Development Strategy adopted in 1971 aspired for an ODA as percent of donor countries GNI of 0.7% target, which is up to date a benchmark aimed for (United Nations [UN], 1971).



ODA, U.S. Dollar billions **ODA, as percent of GNI**

Figure 1. ODA historical perspective 1960-2024 (OECD, 2025a)

By 1960, ODA was around $40 billion. It kept increasing through the cold war era, reaching $90 billion by 1990 (OECD, 2025a). During this period the U.S was the biggest donor accounting for 25% of ODA (Siddiqi, 2005). By the end of the Cold War, ODA declined significantly, shrinking back to $70 billion by 1995. Following the adoption of the MDGs and subsequent pledges to increase aid amounts to respond to these global development goals, ODA picked up in an uprising trend, starting at the range of 1990. The year 2023 witnessed the peak of ODA at $223 billion, almost six folds of ODA by 1960 (OECD, 2025a). The ODA, as percent of donor countries GNI, has fluctuated over the years; generally persisting below the UN set 0.7% target. This is mainly understood in light of the donor countries GNI increasing way faster and beyond the increase in their ODA amounts.

Despite frequent global fora giving promises of increasing aid and sustaining ODA volumes to be responsive to the MDGs and later the SDGs (Cruz, 2023; UN, 2005), aid amounts have been fluctuating and majorly dependent on political and economic situations within donor countries (Greene & Licht, 2018). Starting 2024, there was a remarkable reduction in the amount of aid as per the OECD, which could be understood in light of economic recessions and crisis (Allayarov, 2024; OECD, 2025b). The OECD has reflected that traditional donors like U.S., United Kingdom, France and Germany have all in 2024, for the first time in 30 years, collectively cut their ODA. This cuts trend is expected to continue causing uncertainty on the future of the global agenda (OECD, 2025b). Similar drop in humanitarian assistance in 2024 was estimated to be 11% of previous values (Active Learning Network for Accountability and Performance in Humanitarian Action, 2025). The dissolution of USAID in 2025 and its merger into the U.S. State Department is one of the most discussed shocks in the development and humanitarian aid arenas, given the sudden occurrence and impact on people lives in recipient countries (Edward, 2025; Farber et al., 2025). Simultaneously, emerging donors like China and India have introduced alternative aid paradigms focused on infrastructure and economic cooperation, often blurring the lines between aid and investment (Dole et al., 2021; McCormick, 2008; Paczyńska, 2023). Although some of these shifts occurred after the timeline of the study presented within this thesis, they in fact make this study more important and relevant than ever. It is crucial to examine ways of maximizing the use of the scarce and declining global development aid.

Scarcity in resources, and the above mentioned critiques of aid including the power asymmetry with focus on donors' interests, led to intensive discussion and search for ways to maximize the benefits from the available aid and meanwhile use that to argue for more aid (Barnes & Brown, 2011). After the adoption of the MDGs in 2000, international efforts have been exacerbated to make the best possible outcomes out of available resources since it was clear that increasing fund amount was not sustainably predictable neither sufficient to achieve the intended targets (Deutscher & Fyson, 2008; Vandemoortele, 2011).

As the SDGs are on the horizon as the agreed global priorities within which we "leave no one behind", aid continues to straddle the line between political influence and substantial partnerships. While strategic interests are a recognized driving force, there is still a push toward – and a need for - more equitable and effective cooperation. Understanding this duality—and the evolving context in which it unfolds—is essential for reimagining aid in a way that can cater for recipient countries' agency and development outcomes, especially in the volatile FCAS context. Hence the discussion of aid effectiveness and effective development cooperation become critical.

Aid effectiveness and effective development cooperation

Following widespread critiques and concerns about aid fragmentation and inefficiency, and about the donors' behavior that undermine recipient countries' agency and hence the overall intended development outcomes, the OECD, through its Development Assistance Committee (DAC), initiated and led the process of a global conversation on aid effectiveness in the early 2000s (Dabelstein & Patton, 2013). As a key forum for major bilateral donors, the OECD-DAC provided the institutional infrastructure and convening power necessary to coordinate diverse stakeholders, including donor and partner countries, multilateral organizations, and civil society actors (Bracho et al., 2021). Pursuing the MDGs, the UN organized the International Conference on Financing for Development in Monterrey, Mexico in 2002, leading to the adoption of the Monterrey consensus. The consensus pledged to international cooperation including promises to mobilize more aid for development and reform aid modalities to enable advancing recipient countries systems (UN, 2003). It demanded enhanced aid performance (aid effectiveness) entailing increased harmonization and use of "development frameworks that are owned by recipient countries". The OECD-DAC with the World Bank picked up on

these raised issues (Dabelstein & Patton, 2013) and in 2003 organized the Rome High-level Forum on Harmonization that set commitments that aid should be better harmonized among donors and implementers.

Although some progress was done, following the Rome Forum, in harmonising the work of international aid donors in recipient countries, the aid process was strongly led by donor priorities and administered through donor channels, making it hard for developing countries to lead interventions towards their actual priorities (Dabelstein & Patton, 2013). Further aid was still too uncoordinated and unpredictable with complaints over transparency (Steinwand, 2015). A deeper reform was felt to be essential if aid was to achieve its potential for development reform. The OECD-DAC again called a high-level meeting in, this time focused on how to make better use of aid, producing the Paris Declaration on Aid Effectiveness (OECD, 2005). Over 100 signatories from donor and developing-country governments, multilateral donors, regional development banks and international agencies agreed on a set of five principles for aid effectiveness. They emphasized the importance of moving away from donors' paternalistic tendency to dictate the priorities, toward more *ownership* by recipient countries. Donors and other partners then must *align* their support to the locally defined priorities and *harmonize* interventions among themselves to reduce duplication and loss and foster effectiveness and efficiency. Aid is hence to be operated via system that warrants *management for results* and *mutual accountability*.

The commitments made in Paris were followed in two subsequent high-level fora for aid effectiveness in Accra - 2008 (OECD, 2008) and Busan - 2011 (OECD, 2011) and further refined to address emerging challenges. Yet, the Paris declared principles remained at the core. However, the Busan partnership agreement (OECD, 2011) recognized the rise of new development actors, namely non-traditional donors like China, the increasing importance of non-ODA financial flows for development, and the rising South-South collaboration contrasting the traditional North-South model. Hence, Busan called for a shift from aid effectiveness to the broader concept of effective development cooperation (EDC). It entailed an inclusive partnership framework distinguishing the development actors' multiplicity containing civil society, private sector, philanthropic organizations, and providers of South-South and triangular cooperation. Following Busan, the Global Partnership for Effective Development Cooperation (GPEDC) was established in 2012 (Taggart, 2022), aiming to

advance the EDC agenda monitoring and implementation internationally. The GPEDC today engulfs 161 countries and 56 organization (Global Partnership for Effective Development Co-operation, 2025).

The Paris Declaration (OECD, 2005) and its subsequent sister-agreements marked significant milestones in shifting the discourse toward recipient countries' ownership of priorities, results-based inclusive frameworks with better transparency and mutual accountability. These efforts aimed to transform aid from a top-down imposition into a collaborative process rooted in local priorities and inclusive partnerships (Barnes & Brown, 2011; Deutscher & Fyson, 2008).

Up to date, EDC is rooted in the principles adopted in Paris Declaration on Aid Effectiveness (OECD, 2005). Figure 2 below depicts an indicative framework for the Paris Declaration - from Foresti et al. (2006) - showing how the principles are theorized to achieve development targets through improving countries' system capacity. The framework clarifies how the principles would work within the overall political atmosphere where factors such as partners' pressure, coordination action, political change and disasters affect the outcome and hence the development goals.

It is imperative to note that while Paris declaration focuses on aid as a development tool, it embraces its political underpinnings and relate to the influence of political interests, changes and actions by the various actors.

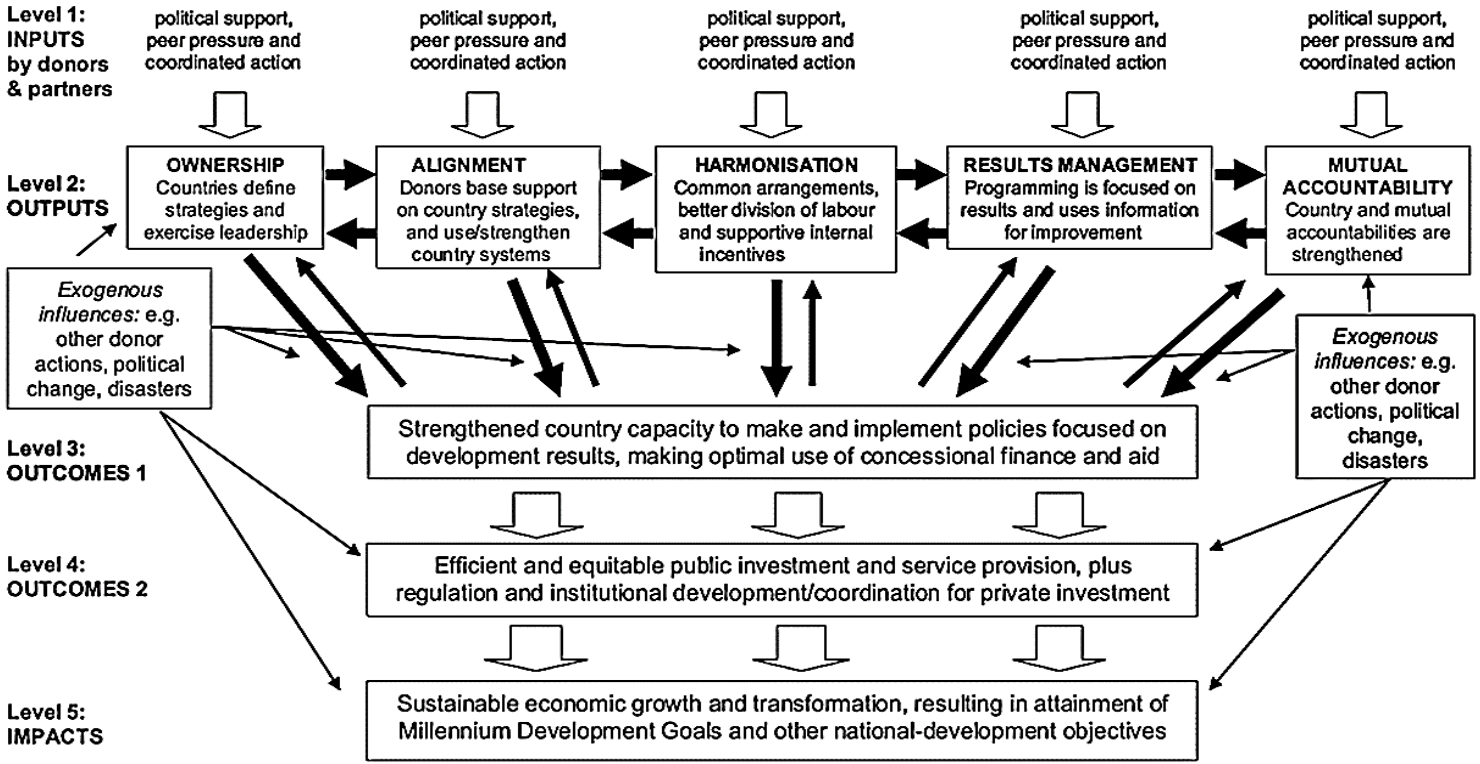


Figure 2. Framework for the Paris Declaration on Aid Effectiveness understanding and evaluation (Source: Foresti et al., 2006)

Health system governance, aid and effective development cooperation

The world health report 2000 (World Health Organization [WHO], 2000) defined health systems as all organizations, people and actions whose primary intent is to promote, restore or maintain health. This definition entails that health systems are purposive towards improving and/or sustaining a good health status among a specific population, via collective actions of people and organizations. The variety of people and organizations that act with different interests and views required some organization within the health systems, which outlines the underpinning of health systems governance. There is a variety of terminologies (including governance, stewardship, leadership) and definitions for what I opted here to refer to as governance. Governance can be understood as the exercise of authority (political, economic or administrative) in the management of affairs within a country (or any demarcation) at the various levels, comprising complex mechanisms, processes, relationships and institutions through which people or groups articulate their interests, exercise rights and duties and arbitrate their differences (WHO, 2011).

Based on these definitions, the World Health Organization (WHO) has developed a list of health system governance functions to guide the exercise of good governance in the health sector (WHO, 2014). These governance functions are:

1. Formulating policy and strategic plans
2. Generating intelligence: information analysis for decision-making
3. Putting in place policy implementation levers and tools – including design of health system organizational structures and their roles, powers and responsibilities; regulation; standard-setting; incentives; enforcement and sanctions
4. Collaboration and coalition-building across sectors and with partners
5. Ensuring accountability via rules and processes for health sector organizations; mechanisms for independent oversight, monitoring, review and audit; transparent availability and publication of policies, regulations, plans, reports, accounts, etc.; and openness to scrutiny.

These five functions illustrate the structural foundations upon which health systems operate and within the various health partners engage. Within FCAS, it is essential to understand how aid for the health sector - Development Assistance for Health (DAH) - relates to these functions.

DAH has steadily increased between 2000-2019, then it witnessed a sharp increase due to the global response to COVID-19 peaking it up to $80 billion in 2021 (Apeagyei et al., 2025). The pattern is similar to what is observed for the overall ODA above. Following COVID-19, a decline in DAH was witnessed in 2023/24 to $49.6 billion, before having an even sharper decline for 2025 due to large donors' aid cuts, projected to end the year at $38.4 billion, a level last seen in 2009 (OECD, 2025b). Even during the two decades of DAH increase, health aid remains unevenly distributed and heavily concentrated in disease-specific programs. By the end of the MDGs era and launching the SDGs, 30% of the DAH was allocated to HIV, 28% to maternal and child health, and only 7% to health system strengthening (Birn et al., 2017c, 2017b). This disease-specific vertical programs focus has been critiqued for undermining comprehensive health system development and for its limited sustainability in fragile contexts.

The landscape of health aid is populated by a proliferating array of actors, including bilateral agencies, multilateral institutions, philanthropic foundations, public-private partnerships (PPPs), and non-governmental organizations (NGOs). The World Bank, Global Fund, Bill and Melinda Gates Foundation (BMGF) and the U.S. President's Emergency Plan for AIDS Relief (PEPFAR), are the largest health donors (Birn et al., 2017b). Each of these actors contribute via distinct modalities such as financial grants, policy support, technical assistance and capacity building (Birn et al., 2017b). Health partnerships scene is dense with continuously emerging partners and mechanisms. For instance, the Roll Back Malaria Partnership alone includes nearly 200 partners, illustrating the scale and complexity of coordination challenges. The proliferation of actors and fragmented funding streams has led to challenges in harmonization, predictability, and transparency. Moreover, donor-driven priorities often overshadow national health agendas, particularly in low-income and fragile states, where aid dependency and limited implementation capacity exacerbate governance and health system challenges (Bogale et al., 2024). Due to these challenges and the critical roles health partnerships play with variation in their support scope and modalities to countries, arrangements to ensure good management of these partnerships are vital. The health sector, early on, embedded the EDC concepts in its strategies and adopted an approach that builds on the Paris Declaration principles, adapting them to specific behaviours for the health partners to change. They were built in the International Health Partnership (IHP+) compact

at its founding in 2007 (UHC2030, 2013). The IHP+ since then served as the global partnership fora to steer aid effectiveness in the health sector, with a secretariat hosted by WHO and World Bank (Alexander, 2007). IHP+ engulfed bilateral donor countries, recipient countries, and international development agencies including UN agencies, Development Banks, Global Health Initiatives like the Global Fund, GAVI and PEPFAR. The membership grew from 26 partners in 2007 to 66 partners in 2016. While the GPED was founded in 2012 to follow the overall EDC progress, the health sector already had such mechanism (IHP+) five years earlier.

The IHP+ adopted seven behaviours including supporting a single national health strategy, harmonizing, aligning and using national financial management, procurement and supply and information system, recording all funds on national budget, south-to-south and triangular cooperation, and providing well-coordinated technical assistance (UHC2030, 2013). They clearly bring the Paris Declaration principles into the implementation domain of health systems, positioning them as inputs toward achieving health development targets. The seven behaviours emphasize three components of the health system as priority areas for strengthening and utilising as part of commitments to align: the financial management, procurement and supply, and health information. Figure 3 below depicts an adapted visualization of how the health system governance functions relate to EDC principles and the seven behaviours.

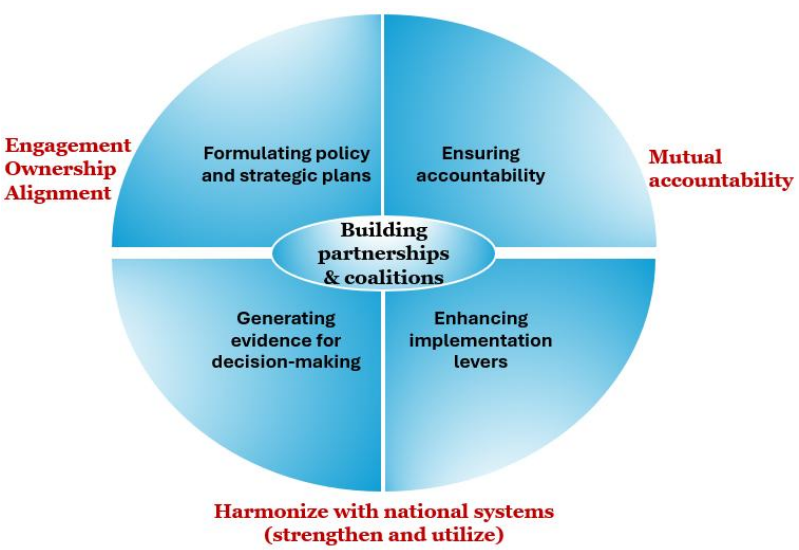


Figure 3. Health system governance functions and EDC
Adapted from WHO (2014) and UHC2030 (2013)

The interlinkages between the health system governance functions and EDC can be easily pointed out once building partnerships and coalitions is centred as the collaboration is an over-arching issue in both. This entails a joint and respectful practice of policy and strategic plan development, that follows principles of ownership and alignment in this engagement. The implementation levers are also addressed within the EDCs call for strengthening and utilizing nationally owned systems and frameworks as part of the alignment and harmonization commitments, including the national health information system as a base of evidence generation for decision-making, hence managing for results. Accountability is indicated in both as a crucial joint process (UHC2030, 2013; WHO, 2014).

Health within the Sustainable Development Goals and effective development cooperation

The SDGs have been celebrated as the Global Agenda to transforming our world by 2030, and were launched by end of 2015 (UN, 2015). Within the SDGs, health was emphasized within a central goal of ensuring healthy lives and wellbeing for all at all ages (SDG 3), with many other health-related SDGs targets. The SDGs witnessed a shift from the disease specific and vertical focus that the MDGs had specially for health goals. Universal Health Coverage (UHC) is the most prominent SDGs health target, with major global commitments towards it by global leaders during the 2019 UN High Level Meeting on UHC (Wright & Mabejane, 2019). UHC entails all people have access to the full range of quality health services they need, when and where they need them without financial hardship (WHO, 2025).

Following the adoption of the SDGs, the global health partners agreed the transformation of the IHP+ into the UHC2030 partnership, keeping the EDC principles in its core but with an UHC focus (UHC2030, 2016b). Additionally in 2019, thirteen leading global partners signed the Global Action Plan for healthy lives and wellbeing (SDG3 GAP) to advance efforts concerning the overall SDG 3 targets (WHO, 2019). The SDG3 GAP signatory partners were health development and humanitarian global organizations, including WHO, World Bank, UNICEF, UNFPA, UNDP, UNAIDS, UN Women, Unitaid, World Food Program, International Labour Organization, Global Fund, GAVI and the Global Financing Facility. Again, within this SDG3 GAP the core principles of EDC are kept as crucial pledges to accelerate progress towards the health SDG targets. Figure 4 below portrays the SDG3 GAP commitments and accelerators.

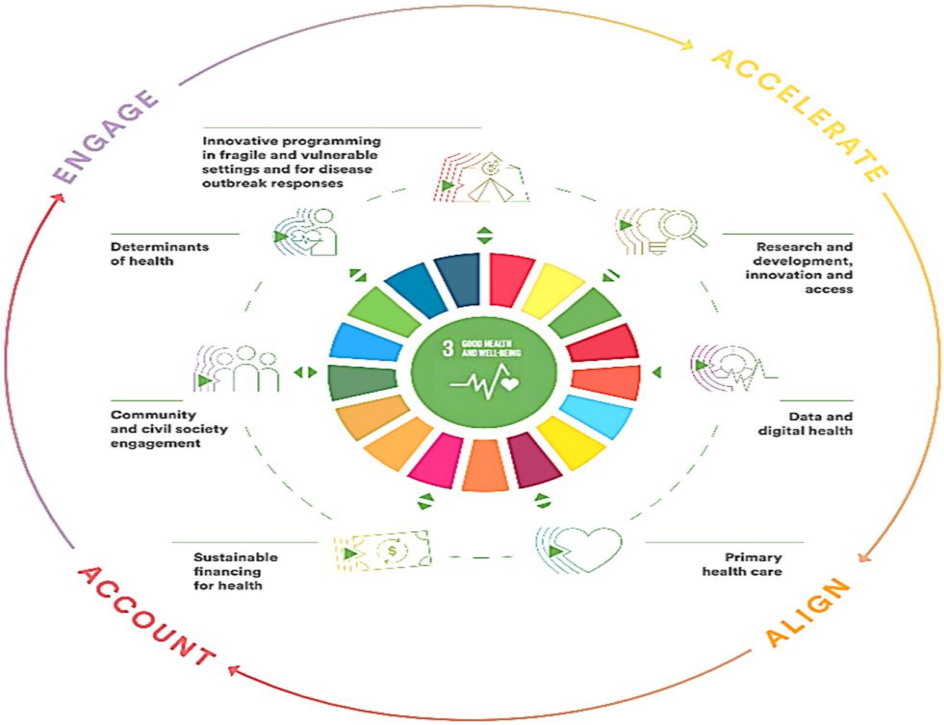


Figure 4. SDG3 GAP commitments and accelerators
Reproduced from WHO (2019) under the Creative Commons Attribution Non-Commercial-Share Alike 3.0 IGO license

Beside classical health concepts and pillars, the SDG3 GAP recognizes that different "innovative" approaches are needed in vulnerable settings like fragile states, especially to respond to emergencies and disease outbreaks. The plan goes to suggest the Humanitarian-Development-Peace Nexus Approach as a policy solution to the witnessed division between humanitarian and development actors. Through the Humanitarian-Development-Peace Nexus Approach various actors from the three sectors (humanitarian relief, development cooperation and peace building) work together in effort to save lives, build local institutions and address the conflict root causes (WHO EMRO, 2021). The variation of humanitarian and development priorities are recognised and yet is believed to be possible to link. The Humanitarian-Development-Peace Nexus approach is hence advocated highly by global actors such as the European Union (EU) and the United Nations (Ramet, 2012; World Humanitarian Summit, 2016). Bogale et al. (2024) indicated that in FCAS the nexus is crucial in ensuring a focus on health system strengthening

and sustainability of services beyond short term emergency interventions through context and phase-based approach. Other scholars also underscore the need to address the humanitarian-development divide and go further to advocate using health as a peace catalyst (Olu et al., 2024). Further, the role civil societies play in advancing health systems is centralized, in line with calls for civil ownership of priorities and active engagement of communities in plans implementation and accountability.

The DAH going to FCAS represented 43% of the overall DAH by 2011, yet a cross-country analysis found that overall aid to low income FCAS was less compared to similar low income stable countries (Graves et al., 2015). This analysis concluded that donors "tended to avoid providing funding to countries with political gridlock, flawed elections, or economic decline". Truppa et al. (2026) call for more stable, flexible and locally aligned DAH in FCAS. Sadly, there is a scarcity of further analysis and research on the external financing for health in FCAS beyond 2011 specially with the changes following COVID-19. The U.S. was the biggest donor to FCAS up to the recent 2024-2025 cuts (OECD, 2025b), and the same applies to DAH in FCAS (Graves et al., 2015). This raises concern about the ability to sustain health services and systems in these settings and calls for further dialogue on EDC. The SDGs commitment to "leave no one behind" places emphasis on reaching the most vulnerable populations, who are mostly in FCAS (World Health Organization & World Bank Group, 2023).

Sudan: a FCAS case study

Sudan is a country located in north-east Africa, bordering seven countries: Egypt, Libya, Chad, Central African Republic, South Sudan, Ethiopia, and Eritrea, and the Red Sea. Its space covers approximately 1.86 million km², making it the third-largest country on the continent. The country has a triangle on the northeast corner disputed with Egypt, outlined with dotted lines in the map below (World Atlas, 2023). The white and Blue Niles runs through Sudan from their origins in Uganda and Ethiopia respectively, and they converge at the capital Khartoum forming the River Nile that runs through the northern region across the Nubian dessert towards Egypt and ends at its outlet at the Mediterranean Sea. In the west, Darfur and Kordofan regions are characterized by semi-arid savanna (Figure 5).



Figure 5. Sudan map (World Atlas, 2023)

Sudan population is around 50 million according to World Bank (2023) among who 36% are in urban areas, with Khartoum alone being the home of 6-7 millions. The predominant language in Sudan is Arabic (World Atlas, 2023), and the majority religion is Islam (Cultural Atlas, 2018).

Sudan gained its independence from British colonization in 1956 (Collins, 2008). Since its independence the country suffered from political instability and internal conflicts. Figure 6 below depicts the major security and political events through the last 70 years of Sudan history.

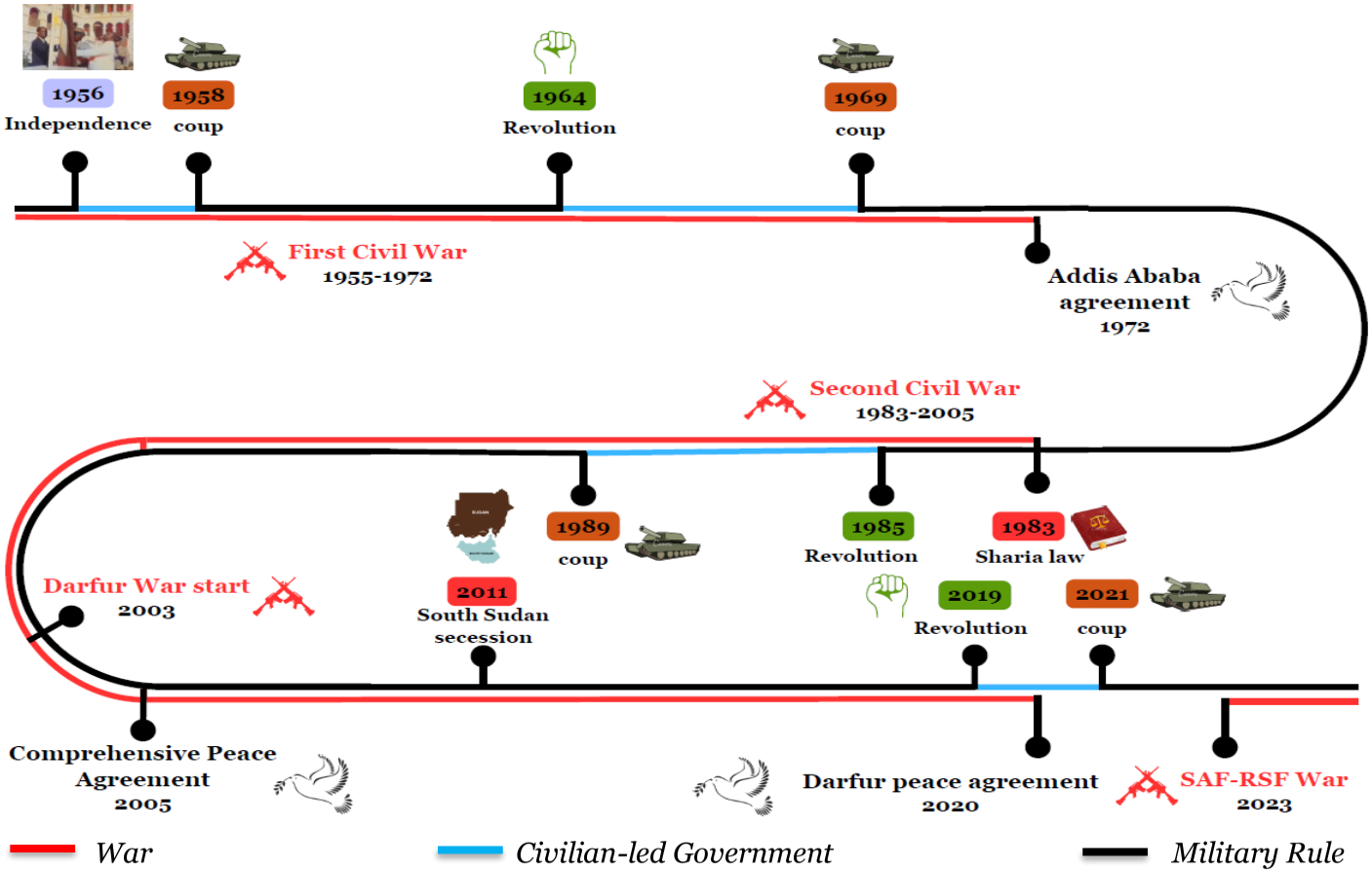


Figure 6. Main political and security events in Sudan 1955-2025

Just months before formal independence, tensions in the south sparked the First Sudanese Civil War (1955–1972). This conflict was rooted in deep ethnic, religious, and regional divisions between the Arab-Muslim north and the African-Christian dominated south, resulting in an estimated 500,000 to 1 million deaths (Johnson, 2003). Meanwhile the political landscape was also unstable, with frequent shifts between civilian and military rule. In 1958, General Ibrahim Abboud led the country's first military coup, initiating a pattern of authoritarian governance that would persist for decades (Collins, 2008). He was forced out of office through a popular revolution in 1964 leading to a brief period of civilian Governance. A coup by Jaafar Nimeiri took over control in 1969. In 1972, the Addis Ababa Agreement ended the First Civil War by granting autonomy to South Sudan, ushering a period of relative calm (Lesch, 1999). However, this peace was short-lived. In 1983, President Jaafar Nimeiri, abrogated the Addis Ababa Agreement and imposed Sharia law nationwide, reigniting tensions and sparking the Second Sudanese Civil War (1983–2005). This war caused approximately 2 million deaths (de Waal, 2007).

In April 1985 again, the Sudanese people took the streets in an uprising that ousted Nimeiri's regime. The transitional military council then handed power to a civilian Government in 1986 (Lesch, 1999). This democratically elected government was also short-lived, and the army in coalition with the National Islamic Front staged a coup in 1989 (Young, 2012). The country was under a single regime control for 30 years, during which the country witnessed major incidences, and continued to suffer deteriorating conditions. The National Islamic Front-led regime drove the country to tensions with many of the international leading powers and organizations, leading to the U.S. aggressive economic sanctions on Sudan (Lyman, 2013). The long-term war in the South reached its end through the Comprehensive Peace Agreement (CPA) of 2005 that led to a referendum and separation of a third of Sudan in 2011 (Young, 2012). Meanwhile, another ethnic conflict in Darfur, west Sudan, issued in 2003 when rebel groups accused the central government of marginalization and the government responded with brutal counterinsurgency, including use of Janjaweed militias (Flint & De Waal, 2008). Human rights violations merited the International Criminal Court (ICC) to issue arrest warrants for Al-Bashir, the regime head, among other leaders (Verjee, 2018). All of this instability caused further weaknesses in Sudan institutions and economic deterioration.

In 2019, a revolution ended Al-Bashir reign over Sudan and was succeeded by a transitional joint military-civilian Government. This transition was again interrupted by a coup from the military leaders who turned on their civilian counterparts in October 2021 (Berridge et al., 2022; Khogali & Homeida, 2023). Shortly after, conflicts within the coup leadership, namely the head of Sudan Armed Forces (SAF) and the para-military Rapid Support Forces (RSF) leader, alongside political tensions after months of demonstrations and negotiations, led in April 2023 to a war starting from the capital (de Waal, 2025). The situation continues deteriorating, with the UN Humanitarian Coordination Forum alarming that Sudan now has the biggest global displacement crisis (United Nations Office for the Coordination of Humanitarian Affairs [OCHA], 2025).

Since the fragile state index inception in 2005 by the Fund For Peace (2025), Sudan has been continuously present within the classification, ranked among the top 10 most fragile countries for 12 consecutive years, highlighting sustained governance and security challenges (El-Affendi, 2023). Similarly, within the World Bank classifications for fragility starting 2006 up to date, Sudan has unbroken fragility classification for nearly two decades (World Bank, 2025). The prolonged periods of conflict with a multiplicity of militias and rebel groups fall within the fragility feature of the loss of a monopoly over the use of force. Further, the repetitive coups and the history of human rights violations lead to erosion of the state legitimacy over repetitive eras, combined with worsening international relations.

On the economic front, political instability, conflict, and changing global support caused significant fluctuations. In the decades following independence, Sudan remained a predominantly agrarian economy with limited industrialization. GDP per capita hovered between US$100 and US$400 from the 1960s to the 1980s, reflecting limited productivity and infrastructure (Suliman, 1998; Wohlmuth, 1989; World Bank, 2022). The discovery and export of oil in the late 1990s and early 2000s led to a period of rapid economic growth, with GDP per capita reaching its peak at approximately US$1,332 in 2011, just before the secession of South Sudan. Sudan's income classification has mirrored these economic shifts. Initially categorized as a Low-Income Country (LIC), during the 2000s oil boom Sudan was reclassified as a Lower-Middle-Income Country (LMIC). Following the 2011 secession of South Sudan resulting in the loss of over 70% of oil revenues, the economy collapsed, and Sudan reverted as LIC.

By 2018, GDP per capita had fallen below US$800, and although there was a brief recovery in 2021 due to transition government global support, the outbreak of war in 2023 caused contraction and GDP shrank by 29% in 2023 (World Bank, n.d.). The economic status along with the conflicts and political instability have limited the state capacity to deliver essential services, with health services as a striking example.

Sudan health system and health partnerships

Sudan's health system is organized as a decentralized, three-tiered model, the federal, state, and locality levels. At the federal level, responsibilities include legislation, policy development, strategic planning, human resource production, technical support, and external relations. The state level is tasked with managing hospital services, employing health personnel, and building capacity. At the locality level, the focus is on delivering primary healthcare services, managing environmental health, and employing assistant health cadres. Despite this structured approach, the system faces significant challenges in resource allocation, service distribution, and institutional capacity (Republic of Sudan Federal Ministry of Health, 2017; WHO EMRO, 2022).

A range of shocks and emergencies frequently disrupt Sudan's health system. These include recurrent disease outbreaks such as cholera and malaria, ongoing armed conflict and political instability, economic crises marked by inflation and currency devaluation, and a fragile infrastructure compounded by the emigration of skilled health professionals. These factors have undermined the health system's resilience and functionality (Health Cluster, 2024; Khogali & Homeida, 2023; Musa et al., 2024; Republic of Sudan Federal Ministry of Health, 2017; WHO EMRO, 2022).

Health financing in Sudan is heavily skewed towards out-of-pocket expenditure, which accounts for approximately 70% of total health spending, primarily directed at private curative services. In contrast, external aid plays a pivotal role in sustaining the health system, particularly in primary healthcare, as it funds up to 75% of the services. This reliance on DAH in the basic and most crucial functions underscores the vulnerability and dependence on international support (Republic of Sudan Federal Ministry of Health, 2015; WHO EMRO, 2022).

A diverse array of partners interplay within Sudan health sector. Multilateral and bilateral donors such as the European Union, Japan, the

African Development Bank, GAVI, and the Global Fund contribute in varying scales to both emergency and development interventions. The Global Fund is regarded as the biggest health sector donor by 2015 according to Sudan Ministry of Health, giving almost $50 million, followed by GAVI which provided a support in the range of $30 million. The European Union provided slightly above $5 million by then, while the African Development Bank did not exceed $1.128 million during that era (Republic of Sudan Federal Ministry of Health, 2016). UN agencies including WHO, UNICEF, and UNFPA are central actors, often balancing implementing humanitarian and development mandates, with UNICEF presenting the biggest fiscal portfolio among UN agencies to the health sector at around $41 million in 2016, followed by WHO at $30 million. International and national NGOs predominantly focus on humanitarian service delivery, although some have expanded into development roles. Civil Society Organizations (CSOs) are increasingly involved in implementing projects financed by global partners. CSOs have been defined as national NGOs, professional associations and academia working in the health sector in Sudan (Republic of Sudan Federal Ministry of Health et al., 2014). Through CSOs informants, I have found that the CSOs have formulated a network for organizations working in health that is mainly mandated to organize their involvement in implementation of major grants (e.g. representation in the Global Fund country coordination mechanism) and capacity building exercises.

Some donors like World Bank and USAID have significantly picked up on health sector financing after the revolution. The World Bank COVID-19 health system response fund was $120 million for 2020-21 (World Bank, 2020, 2021), while USAID health emergency fund exceeded $70 million for those years (U.S. Agency for International Development, 2020, 2021).

A variety of implementation modalities have been adopted by the different partners in Sudan health sector for different reasons. Western donors like the EU and USA rely only on UN agencies and International NGOs, with national CSOs as secondary implementers, and no implementation through Government entities (Republic of Sudan Federal Ministry of Health, 2016). The Global Fund in Sudan has been implementing the additional safeguard policy since 2005 to protect the use of its resources by implementing extra risk mitigation measures (The Global Fund, 2015). This has led to the assignment of UNDP as the principal recipient of the Global Fund grants, a status that continued until the Ministry of Health

started being a principal recipient for the health system grant from 2016 and Malaria grant in 2017. Yet, HIV and TB grants have endured being handled by UNDP, and in 2024, the Malaria grant principal recipient role has shifted to UNICEF (The Global Fund, 2026). The Global Fund has provided support via Projects Management Units (PMUs) hosted by grants principal recipients. Similarly, GAVI has mostly implemented vaccination interventions through UNICEF, while having their health system grant implemented through the Ministry of Health managed by a specific PMU. The UN agencies also have deployed a variety of implementation modalities. While classically they went for direct implementation specially for humanitarian interventions or upon donors' conditions, they also have had a direct financial contribution modality where the government receives funds as per agreed annual plans following a request process. This later approach was enhanced by the operation of a Harmonized Approach to Cash Transfer (HACT) where the Government receives funding for activities, implement and carry all the reporting and financial procedures, and the UN agencies held the right for spot checks and audits. The International NGOs have carried out mostly humanitarian interventions, solely relying on direct implementation in the ground. Sometimes INGOs even shouldered the responsibility of managing health services in conflict-affected geographical areas with weak Government presence (Republic of Sudan Federal Ministry of Health, 2016, 2017).

To enhance coordination among this wide array of actors, address the reducing resources for health - both domestic and external - following South Sudan separation, and improve aid effectiveness, Sudan has adopted several partnership mechanisms and frameworks. The Sudan Health Compact, signed in 2014 (Republic of Sudan Federal Ministry of Health et al., 2014), represents an agreement among key stakeholders (relevant government ministries and authorities, donors like USAID and EU, African Development Bank and World Bank, UN agencies, INGOs and CSOs) to align with national priorities and strengthen the national system.

Sudan's participation since 2011 in the IHP+ reflects an enthusiasm to the EDC (Republic of Sudan Federal Ministry of Health, 2017). The adoption of the HDPNx as a strategic shift within the health policy in 2017 aimed to bridge the divide between emergency and development responses. Additionally, the Health Sector Partners Forum, established in 2016, serves as a coordination platform under the government leadership that brings both humanitarian and development actors (WHO EMRO, 2022).

Despite all these agreements, approaches and platforms, EDC implementation in Sudan faces instability and contextual complexities. These complexities highly affect the health system governance in an already fragile state.

Study rationale

FCAS face profound and persistent challenges in achieving health system and development goals due to weak governance, fragmented systems, recurrent shocks and chronic underfunding (Landry et al., 2021). These conditions undermine national capacity to plan, finance, and deliver essential services, making FCAS disproportionately dependent on external assistance (Bogale et al., 2024). This amplified calls to rethink how development partners engage in FCAS and how national institutions can be strengthened to ensure sustainable progress toward development goals (Carment & Samy, 2023). Against this backdrop, EDC has gained prominence as a framework to overcome long-standing critiques of aid—namely, power asymmetries, fragmented donors' interventions, and bypassing of state systems—that have hindered aid's sustainable development impact in recipient countries (Barnes & Brown, 2011; Carment & Samy, 2023; Cruz, 2023; Deutscher & Fyson, 2008). Empirical evidence on how EDC principles function in FCAS, and their actual effect on systems strengthening and development outcomes, remains limited (Carment & Samy, 2023; Zürcher, 2022). There is a broad scarcity of understanding EDC implementation in reality and its impact on the overall development targets (Gisselquist et al., 2024; McKee et al., 2020).

The health sector in FCAS is generally characterized by institutional fragility, parallel service delivery structures, and a high concentration of humanitarian actors whose mandates and time horizons often diverge from long-term development needs (Saez et al., 2021; Spicer et al., 2020). DAH in such contexts overwhelmingly supports vertical, disease-specific programs, with only a small share directed at overall system strengthening (Birn et al., 2017b). As a result, national systems frequently remain under-resourced, underutilized, and overshadowed by partner-driven mechanisms, perpetuating fragmentation and weakening governance. This has prompted global health partnerships such as the IHP+ and the SDG3 GAP to explicitly embed EDC principles within partners practices, emphasizing alignment with national plans, use of country systems, coordinated support and joint monitoring (UHC2030, 2013; WHO, 2019).

Yet, there is a critical gap in studies that investigate EDC in practice (Haque et al., 2017; Lokot et al., 2022; Truppa et al., 2024). Consequently, there are limited insights on whether, how, and under what conditions EDC can contribute to strengthening health systems, governance and improving health sector outcomes in FCAS.

Sudan represents a compelling and highly illustrative FCAS case through which to study these EDC and health system governance issues. The country has experienced decades of conflict, frequent political shifts, and donors sanctions with contestations of the state legitimacy, all of which have shaped its health system and external financing landscape (de Waal, 2025; El-Affendi, 2023; WHO EMRO, 2022). Sudan has formally signed EDC agreements, including the IHP+ global compact and the nationally endorsed Sudan Health Compact (Republic of Sudan Federal Ministry of Health et al., 2014) and adopted the HDN approach in 2017 (WHO EMRO, 2020). Sudan therefore offers a unique opportunity to analyse how contextual volatility interacts with partners' behavior and governance processes in shaping the practice of EDC.

This doctoral thesis seeks to address some gaps that the literature on EDC underscores. First, while the EDC principles have been widely promoted, their operationalization in fragile settings remains poorly understood—particularly how partners interpret and apply principles such as ownership, alignment, and harmonization in contexts of contested legitimacy and weak institutions (Carment & Samy, 2023; Lokot et al., 2022). The evolvement of the partners relations and practice of the principles within Sudan frequently changing context will enrich the understanding of this issue.

Second, although the Humanitarian–Development Nexus approach has been heavily promoted as a central global strategy (new way of working) in FCAS, rigorous empirical evaluations of its implementation and effects on system strengthening are scarce, especially with documented humanitarian-development actors differences (German Federal Ministry for Economic Cooperation and Development, 2021; Lie, 2020).

Third, while the national health system strengthening and utilization of unified system and arrangements is widely discussed and considered the backbone of health system governance and are explicitly prioritized in global EDC frameworks (UHC2030, 2013; WHO, 2019), the literature highlights limited empirical evidence on how partners engage with—and

either strengthen or bypass— the national system components such as public financial management, procurement, and health information (Bogale et al., 2024; Witter et al., 2020).

Lastly, although EDC principles are theorized to contribute to health and development outcomes, there is a lack of comparative, cross-country analysis on whether these principles are actually associated with achieving these targets in FCAS (Gisselquist et al., 2024; Truppa et al., 2024).

Taken together, these gaps demonstrate the need for a comprehensive, empirically grounded examination of EDC in FCAS that spans policy interpretation, partnership dynamics, system governance, and health outcomes. By combining a country case study of Sudan with a comparative cross-country analysis, this doctoral research provides timely and relevant insights into how EDC can serve as a lever for resilient health systems and sustainable development in fragile contexts.

Aim and objectives

The overall aim of this doctoral study was to examine the effective development cooperation efforts, challenges, and opportunities in influencing health system governance in fragile and conflict-affected states, toward achieving UHC.

The project has four objectives that have been covered in 4 sub-studies within the following order:

1. To analyse the evolvement of Sudan health sector partners' relations, perspectives, and adherence to the EDC principles of ownership, alignment, and harmonization during the period 2015–22 (paper 1).
2. To explore the understanding and adoption process of the Humanitarian-Development Nexus approach to health system strengthening in Sudan (paper 2).
3. To investigate the evolving attitudes and practices of Sudan's health sector partners towards strengthening and utilizing the national health system components of financial management, procurement and supply, and health information (paper 3).
4. To examine if and which EDC principles have affected the attainment of SDG3 targets in aid recipient FCAS, focusing on UHC (paper 4).

Theoretical framework

Building on Paris Declaration evaluation framework (Foresti et al., 2006), the IHP+ seven behaviours for partners to change (UHC2030, 2013) and the SDG3 GAP theory of change (WHO, 2020), a theoretical framework was developed in a theory of change format. It is outlined in Figure 7 below, with the four sub-studies positioned within it.

The theory of change outlines that applying the aid effectiveness principles (Paris Declaration as the core) as inputs will mean that aid is delivered through and catalyses better coordinated, aligned and effective approaches. Among these approaches, the HDN stands out as a special approach of relevance for FCAS. These approaches represent the change pathway towards achieving intermediate outcomes and results. The main theorised intermediate outcome is strengthened countries' capacity to implement properly jointly developed policies that correctly defines priorities, focus on results and optimally use available resources. The overall expected results are the sustainable development targets, namely for the health sector defined as UHC among other SDG3 targets. The inputs (principles), change pathway, intermediate outcomes and results are expected to be all influenced by, and interact with, a set of factors encompassing political environment, partners pressure, coordination action and stability vs disasters within the overall context. These are conceptualised as main internal and external influences that could be seen as opportunities to utilize or risks and barriers to mitigate and overcome towards enabling the theory of change to yield the results.

Within this theoretical framework, sub-study 1 analyses the principles of ownership, alignment and harmonization, examining partners' behaviour and relations, their evolving and interaction with the changing contextual factors. Sub-study 2 explores HDN as a change approach and examines its understanding and adoption process. Sub-study 3 investigates the partners' attitudes and practices towards strengthening the country's health system capacity and utilizing it (principles as inputs toward intermediate outcome) and their evolution within the changing context. It focuses on the three system components of financial management, procurement and supply, and health information system as priorities. Within these first three sub-studies the selected case of study was Sudan. Sub-study 4 examines if and which EDC principles have affected the attainment of results (SDG 3 targets, centring UHC) in aid recipient FCAS.

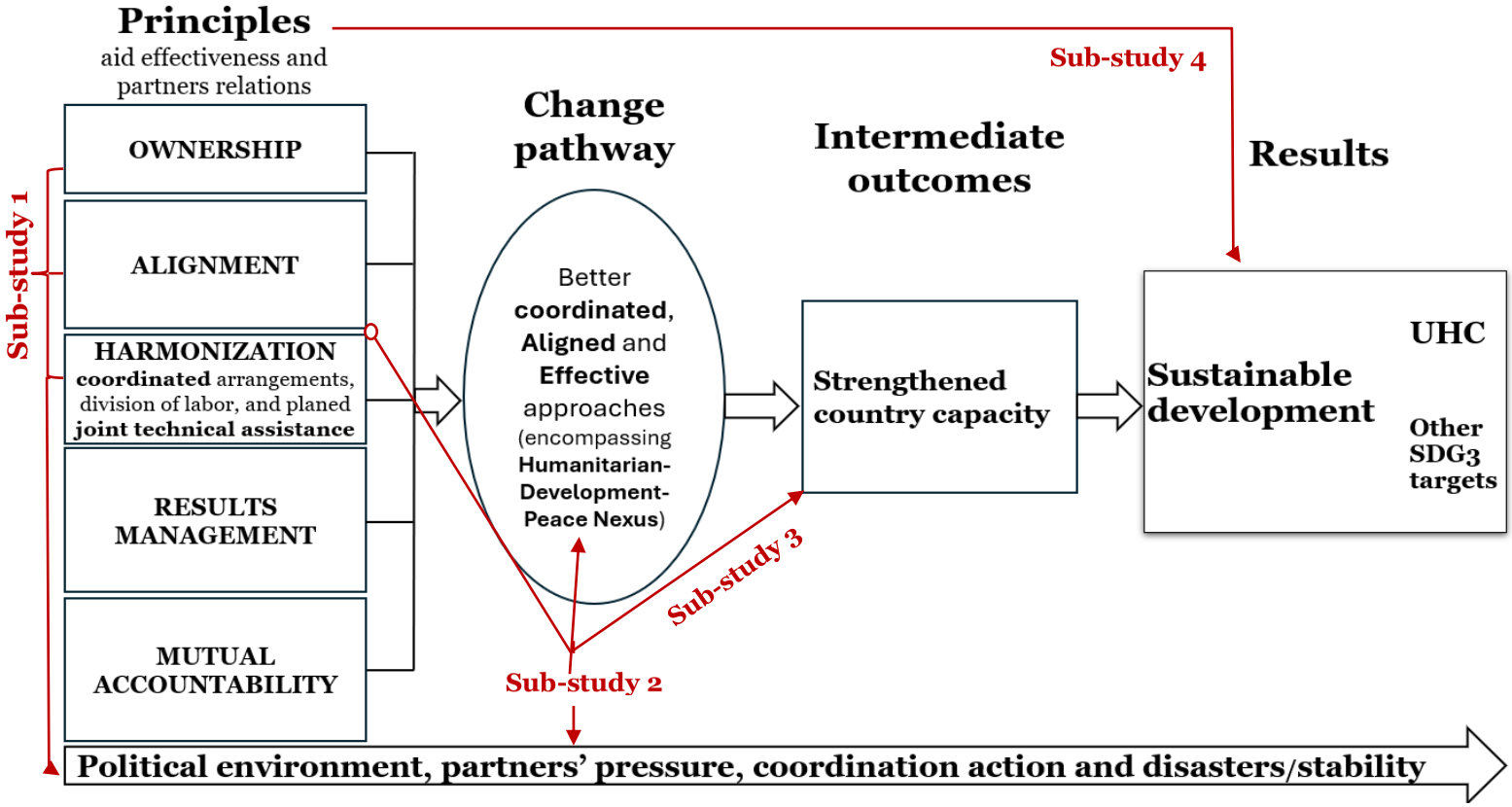


Figure 7. Theory of change: implementing effective development cooperation towards health development goals (adapted from Foresti et al., 2006; UHC2030, 2013; WHO, 2021) with sub-studies positioned

Materials and Methods

Design

I applied both qualitative and quantitative approaches to achieve the study objectives. In the first 3 sub-studies, I implemented a qualitative case study design as the aim was mainly exploration and understanding of perspectives of key health partners within the specific context of Sudan. In the fourth sub-study I opted to a quantitative approach (qualitative comparative analysis) as the objective was examining if and which EDC principles have affected the attainment of UHC in several FCAS. This thesis tries to integrate the three qualitative sub-studies and the quantitative sub-study to provide nuanced comprehensive understanding of how EDC is practiced, how it affects health system governance and impacts the attainment of health targets.

Table 1 below highlights the details of the four sub-studies in terms of design, data, participants and cases, and analysis methods.

Data collection

The data for the qualitative studies (sub-studies 1, 2 and 3) were mainly obtained via semi-structured interviews with key stakeholders of relevance, supplemented by documents review. Further, coordination meetings observation notes were utilized to triangulate the findings. The data for these sub-studies were collected in two time periods, the first in 2015 and the second in 2021 and 2022, corresponding to different contextual situations and underpinning EDC commitments and practices. The data for the quantitative study was obtained from publicly available sources (secondary data). I further describe the data collection process, instruments, participants and cases below.

Figure 8 below positions the data collection within a timeline of the main political and EDC relevant events in Sudan for the first three sub-studies. It is important to note that for the second data collection round (2021/22), I conducted interviews with two different semi-structured guides and sets of participants, of which 8 interviews fed into sub-study 1 and 7 interviews into sub-study 2; while all 15 interviews provided data for sub-study 3. Sub-studies 1 and 3 used the first-round interviews conducted in 2015. Sub-study 2 only relied on the 7 interviews of relevance during 2021/22.

Table 1. Summary of the methodology per sub-studies

	Sub-study 1	**Sub-study 2**	**Sub-study 3**	**Sub-study 4**
Design	Qualitative case study	Qualitative case study	Qualitative case study	Quantitative (comparative study)
Data collection	Interviews (16), meetings observations, and documents review	Interviews (7), supplemented by documents review	Interviews (15), supplemented by documents review	Secondary data (publicly available)
Participants / cases	- Sudan government - Donors and Development Banks - UN Agencies - International NGOs - Civil society organizations	- Sudan government - Donors - UN agencies - International NGOs (both humanitarian and development constituents)	- Sudan government - Donors and Development Banks - UN Agencies - International NGOs - Civil society organizations	Aid recipient FCAS (17)
Instruments and data sources	- Interview guide - Memos + notes from interviews + documents + observations.	- Interview guide - Memos + notes from interviews + documents + observations.	- Interview guide - Memos + notes from interviews + documents.	- UN SDGs monitoring - UHC tracking
Analysis	Framework analysis	Thematic analysis (policy triangle)	Situational analysis + positional maps	Qualitative Comparative Analysis
Status	Published	Published	Submitted	Submitted

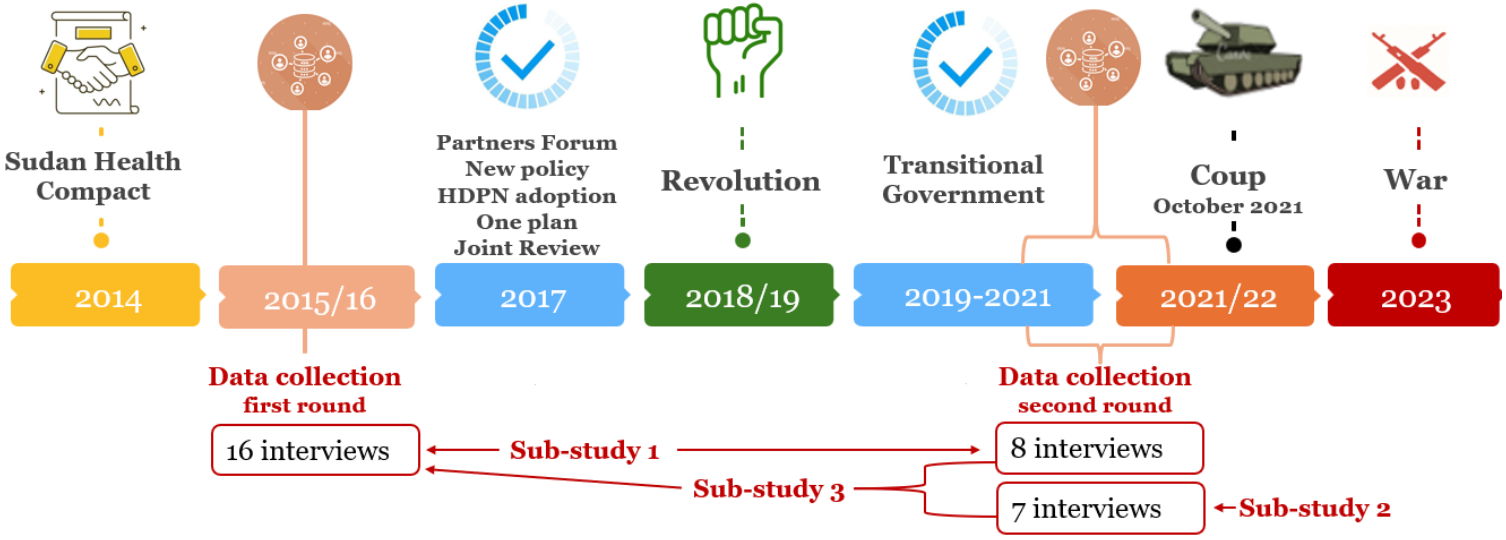


Figure 8. Data collection timeline – Qualitative studies

Qualitative data collection – first round (2015)

The qualitative studies relied mainly on informants' semi-structured interviews; supplemented with notes of observations for two coordination meetings and review of relevant documents (spanning agreements, strategies and plans). The data in 2015 were part of my master thesis.

During July and August 2015, 16 in-depth interviews were conducted. The target were senior technical staff, who were directly involved in health sector partnerships and health system deliberations. Initially I conducted a quick scan and analysis of stakeholders and developed a list of potential participants to cover different constituencies. The final list included high-level staff from the following partner categories:

- **Sudan government:**
  Refers to the public authorities responsible for governing the country and overseeing public policy, including ministries, national agencies, and sub-national governance structures, with the Federal Ministry of Health as the leader in this case.
- **Donors and development banks:**
  Includes bilateral donors (e.g. USAID, FCDO), multilateral donors (Global Fund, GAVI), and international financial institutions (World Bank, African Development Bank, Islamic Development Bank), all of which provide mainly financial aid accompanied (or not) with technical and/or policy support to Sudan.
- **UN agencies:**
  The main United Nations agencies functioning within the country in support for the health sector (e.g. WHO, UNICEF, UNFPA …etc). Their mandate is mostly provision of technical expertise and guidance, as well as facilitating implementation of donors grants in some cases.
- **International NGOs:**
  Non-governmental, non-profit organizations operating worldwide to deliver health services through implementing humanitarian aid and development assistance (e.g. IRC, MSF, Save the Children, Plan …etc).
- **National Civil Society Organizations (CSOs):**
  Locally established, non-state organizations, including community-based organizations, advocacy groups, faith-based groups, and professional associations. They excel on community mobilization, rights advocacy, contextual delivery and trust building.

Potential interviewees from the different entities were identified based on being active senior technical staff from partner entities in Sudan health sector within the definition of partners as per the Sudan Health Compact (Republic of Sudan Federal Ministry of Health et al., 2014). A list of health partners was acquired from the Directorate of International Health, Sudan Ministry of Health, and the WHO office in Sudan, and further welcomed suggestions from respondents during the interviews. The final list had 24 potential interviewees who all were contacted. Interviews were conducted with the 16 who responded positively. All interviews were conducted in participants' natural settings (mainly offices) as I believe their natural work setting would help improve their sense of comfort and consequently facilitate the free exchange of information.

A semi-structured interview guide based on the EDC commitments with a focus on the three principles of ownership, alignment, and harmonization was utilized and mainly fed sub-study 1, while having specific questions on system strengthening and utilization that fed into sub-study 3 (see Annex 1). The guide was amended when found necessary as the revision of the conducted interviews and the notes inspired some necessary changes from arising issues up to the attached version. All interviews were conducted in English as all respondents were fluent in it.

Other data sources (observations and documents) were used to confirm and complement the findings and ensure validity and reliability, thereby making it possible to check partners' inputs thoroughly. I conducted observations with notes taken for two coordination meetings during the same period. Additionally, documents that were found relevant to the study or were suggested by interviews participants were compiled.

Qualitative data collection – second round (2021/22)

For the second data collection round (May 2021–May 2022), interviews were conducted with relevant respondents from the constituency groups that Sudan Health Compact defined corresponding with the same constituencies' groups in the first round. We followed the same process from the first round, starting with contacting respondents from the first round who were found still to be working in Sudan. Saturation for data utilized for sub-study 1 was reached after 8 interviews, while it took 7 interviews to reach saturation for sub-study 2. Sub-study 3 utilized all 15 interviews from this round. All interviews were conducted in English as

all respondents were fluent in English language. Further, documents mentioned by respondents or found of relevance were utilized.

For sub-studies 1 and 3, I used the same interview guide from first round with a lens of follow-up on changes in context, perspectives and actions.

For sub-study 2, a second semi-structured interview guide was developed based on the second objective (see Annex 2). The interview guide addressed the issues of the nexus understanding, introduction, and implementation process, as well as its implementation progress, challenges, and opportunities. While exploring these aspects, the policy aspects of the content (the Nexus concept), actors, process and context were studied. This guide also included questions on the health system strengthening and utilizations perspectives, similar to those on the first guide, so these seven interviews were also used for sub-study 3. The interview guide was frequently amended during the study process. The respondents included 4 of the above 5 constituency groups, excluding CSOs due to the lack of their involvement in the nexus adoption in Sudan health sector up to the point of our data collection.

For this round, all the interviews except 1 were conducted via online Zoom calls, as Zoom was widely used for partner meetings in the Sudanese health sector following coronavirus disease 2019 (COVID-19). All respondents were of high rank with good alternative internet network arrangements. Interviews were recorded with respondents' consent.

Four respondents were common in both data collection rounds.

Quantitative data - sub-study 4

The data for the quantitative sub-study was obtained from several publicly available sources (secondary data). The study focused on aid recipient, fragile states, all within a comparable economic status (LICs and LMICs). This was intentional as UHC is subject to complex interactions of political, economic and social factors, so cases of as homogeneous nature as possible were preferred to enable the examination of the relation between EDC and such outcome. The initial list of cases was obtained from the SDG3 GAP monitoring report 2021 (WHO, 2023) which included 42 aid recipient countries and territorial states. The list was filtered to include only countries that were classified as fragile states (World Bank, 2025) and as either LICs or LMICs by World Bank (World Bank, n.d.) in 2021.

Finally, we only included cases for which the outcome variable of UHC index was available, yielding a total of 17 cases.

The study variables included all the six conditions (principles) of EDC that the SDG3 GAP monitoring cycles report on (WHO, 2023). These are listed below, with the used abbreviation in bold.

1. The partners **align** with the national plans.
2. **Partners coordinate** their support with each other.
3. Using a unified **national coordination** mechanism (no parallel mechanisms).
4. Partners' financial support is in line with national **budget** priorities.
5. Partners use national monitoring and evaluation **(national M&E)** system and report within it.
6. Partners have a joint technical assistance (**joint TA**) plan.

This data was gathered by the SDG3 GAP secretariat. The secretariat collects the information from SDG3 GAP focal points at national governments, reporting on GAP partners' performance on these EDC principles in the form of a rating scale from 1 to 5 (strongly disagree to strongly agree). This is done through a questionnaire circulated annually.

For the outcome, I chose the UHC service coverage index due to its centrality as the main health target within the SDGs. The data for this index was retrieved from the tracking UHC global monitoring published by the World Bank and World Health Organization (World Health Organization & World Bank Group, 2023). Data was available in percentages, with the index representing an average coverage of defined essential services including reproductive, maternal, newborn and child health services, non-communicable diseases and infectious diseases service capacity and access (World Health Organization & World Bank Group, 2023). We utilized the index data for the year 2021, which was the most recent data available during this study conduction time (2025). The full data set, per the 17 included countries is enclosed in annex 3. Further, I collected data for the under-five mortality rate (U5MR) from the UN SDGs data base (2025), and utilized it to run a supplementary analysis with U5MR relative change as the outcome. This indicator was chosen due to its relevance as a classical health outcome indicator with readily available data. Details on this supplementary indicator and analysis are provided in annex 4.

Analysis

This section presents the analytical strategy utilized per each sub-study. For the qualitative sub-studies, the initial step was transcription. The recorded interviews were transcribed verbatim, revised by listening back to recordings and rereading the transcripts to ensure accuracy and familiarize more with the data. Notes and short memos were incorporated into this process.

Sub-study 1

For sub-study 1, the framework analysis approach was applied. This framework technique is defined by Gale et al. (2013) as "a set of codes organized into categories that have been developed by the researcher(s) that can be used to manage and organize the data." The codes are descriptive or conceptual "labels" assigned to the excerpts of raw data. The codes were then entered into the framework matrix, a spreadsheet containing numerous cells into which summarized data are entered by codes (columns) and cases (rows, representing the respondents, allowing comparison of inputs within the two datasets). Themes were then developed as interpretations or propositions to describe and explain aspects within the data, i.e. interrogating data categories through comparison between and within cases. This allowed careful comparison not only between participants' inputs but also within the same participant and mid constituency representatives over our two data collection rounds. A combined approach to analysis was utilized, enabling the development of themes both inductively from the accounts of participants and deductively using the framework for the Paris Declaration commitments for understanding and evaluation (Foresti et al., 2006).

Using OpenCode 4.03 (ICT Services and System Development & Division of Epidemiology and Global Health, 2015), I conducted a line-by-line coding of the interviews. Initially, beside myself as an insider who carried out the interviews and was familiar with the context, one outsider (Daniel Tsang) assisted the process of open coding. Due to the feedback received from him on the unclarity of how I reached some codes, often, we agreed to use in vivo codes as "Owner of the house has the responsibility to coordinate" and "If Ministry of Health wants a pink building, we do it pink". Comparing the different codes and reflecting on these and the memos inspired the grouping of codes into categories in a deliberate

process, going forward and back, revising the categories and the fit of the codes on each category and their relevance to the data.

For the rest of the interviews, I used more selective coding based on the Paris Declaration framework, with a focus on the three principles of ownership, alignment and harmonization, guided by the open coding outlines. I continued the process of coding (systematically applying coding labels, identified by the framework method as indexing in a deductive manner) while trying to keep attention to recognize emerging codes and issues. Hence codes emerging from the transcripts were added to the now deductively decided list of codes.

I then grouped coding labels together conceptually into categories (using a tree diagram), which changed over time, becoming successively clearer through coding additional interviews. This formed a working analytical framework that I only closed after analysing all the data. The final framework contained 58 coding labels grouped into nine categories. Microsoft Excel spreadsheets were used to chart the data into a matrix. The cells included quotations and texts under each coding label (column) for each respondent (row). Separate sheets were used for each category. Examining patterns within each group of coding labels and comparing the responses of different interviewees and the different partner categories assisted in interpreting the analysis framework matrix. Themes emerged based on these patterns, assisted by the memos and notes, as well as comparing them with observation notes. In this process, the focus was on going beyond simple descriptions to explanations of differences and the reasons behind attitudes and acts linking the different parts of responses.

The second set of data had a more focused analysis, comparing the data with the first-round analysis and building on it, while keeping an eye open for any issues arising. The focus was on defining any changes from 2015 to report and relate them to the policy and contextual changes for Sudan (revolution then coup) and the health partners relations and commitment.

Sub-study 2

A reflexive thematic analysis was adopted for this sub-study following Braun and Clarke (2021) guidance. Utilizing OpenCode 4.03 (ICT Services and System Development & Division of Epidemiology and Global Health, 2015) line-by-line coding of the transcripts was performed inductively. Codes were compared and reflected on against the sub-study aims and

written memos and subsequently grouped into initial sub-themes. This was fostered by rounds of revising the initial sub-themes and the codes' fit on each and their relevance to the data. Then, examining patterns across sub-themes helped formulate themes. The themes were further revised and organized using the policy triangle framework (Walt & Gilson, 1994).

Sub-study 3

Analysis for this sub-study applied the situational analysis technique with positional mapping (Clarke, 2005; Clarke et al., 2015). Coding process of all the transcripts was performed via OpenCode 4.03 (ICT Services and System Development & Division of Epidemiology and Global Health, 2015). The codes were then reviewed with constant comparison within and with the original text. Codes containing similar ideas regarding system strengthening or system utilization were grouped together into categories.

These codes categories were revised against a matrix of summarized texts of respondents' inputs on the system components and their status, strengthening efforts, and utilization stands, to ensure their accuracy. Finally, a positional map was constructed as a visual representation of the final categories. This process was first done separately for data from 2015 and data from 2021/2022 leading to two separate maps for each data set. Positions from the two maps and the specific inputs of partners within the two data sets were compared, enabling merger of the maps into one that depicts both common positions and positions that were found within only one data set (Figure 9). This allowed visualizing the persistent attitudes and practices, while also observing changes over time.

Two axes were used to map the positions, one capturing the stands on system utilization (the x-axis), and the other which captures stands on system strengthening (y-axis). The positions within the map depict the respondents' input place within a range from a rejection stand to a strongly supportive view or action. For example, the scenario where a partner strongly supports utilizing the national system can be seen with statements like "We contracted NMSF for our supported facilities". A stand of "we started supporting the national system so that we can be able to use it in the future" is seen as supporting health system strengthening and being open towards utilizing it but with reservation at current time. When a respondent says "We don't follow this common system idea" then this is taken as rejection for both strengthening and utilization intents.

Sub-study 4

Analysis for this sub-study was carried out through the Qualitative Comparative Analysis (QCA) method (Bazzan, 2023; Mello, 2021; Rihoux & Ragin, 2009). QCA uses the strengths of Boolean algebra and inferential statistical reasoning to investigate the necessary conditions (for the outcome to happen they must be present) and the sufficient conditions (whenever present the outcome happens). QCA gives solutions that outline causal pathways and measures of fitness.

Calibration is the first central step in QCA, where the data is transformed from its original raw numerical format to membership scores based on a theory-based threshold. I applied a crisp-set calibration, meaning that I dichotomized the variables (both conditions and outcome) into in and out, i.e. success or failure in meeting threshold, based on theoretical criteria. Given that the assessed EDC conditions all followed a 5-point Likert scale, ranging from strongly disagree to strongly agree, I opted to consider acquiescence bias. Acquiescence bias is when self-rating respondents tend to agree with statements as default, especially when they are unsure, leading to more frequent selection of "agree" rather than "strongly agree," (Kam & Meyer, 2015). Hence, only "strongly agree" was considered here as presence of the condition.

The tracking UHC progress report (World Health Organization & World Bank Group, 2023) categorized the UHC index data into five quantiles: very high (>80), high (60-79), medium (40-59), low (39-20), and very low (<20). I opted for a threshold value of 40 or above for the outcome present based on this rationalization. Annex 3 shows the data before calibration.

Once the data was calibrated, an initial necessity and sufficiency analysis was carried to assess the pathways to the outcome. Two parameters of fit were applied: ***consistency*** (proportion of outcome set that is included in/part of the condition(s) (set) of interest) and ***coverage*** (the empirical relevance of the relationship i.e., how many cases are explained by this relationship); to assess the goodness and the strength of the set relations. I applied a consistency threshold for necessity of 1, and 0.75 for the sufficiency, as instructed by Rihoux and Ragin (Rihoux & Ragin, 2009). The analysis of necessity and sufficiency was conducted for both presence and absence of the outcome and the presence and lack of the conditions.

Following this step, the truth tables, i.e. tables that list all the possible combinations of the conditions and examine within which combinations

the cases fall, were created. This entails the truth tables also show combinations of empty cases, called logical remainders. To simplify these tables, I applied a minimization function using the parsimonious solution. Minimization uses Boolean algebra to reduce truth tables to a simpler set of causal pathways that explain the outcome. The parsimonious solution eliminates logical remainders with contradictory simplifying assumption (logical remainders with configuration of conditions sufficient for the outcome and non-outcome simultaneously) and logical remainders that contradict the claims of necessity (logical remainders that have the outcome without the claimed necessary condition, in case necessity relation claimed). The parsimonious solution then shows possible model(s) of the pathway to the outcome(s) with measures of consistency and coverage, as well as the covered cases within each pathway model.

Further coverage concepts in the format of unique and raw coverage were produced at this stage. Raw coverage (RCov) refers to how much of the outcome is explained by conditions set, while unique coverage (UCov) refers to how much of the outcome explanation can be attributed only to that set / condition and not any other.

The analysis was conducted using the packages QCA and SetMethods within the R program (Bazzan, 2023; Duşa, 2024).

Ethical considerations

Ethical clearance

For the qualitative sub-studies, an administrative letter was issued from the designated directorate at Sudan Ministry of Health for data collected in 2015, as part of a master thesis study back then. This was the only possible approval due to the non-functional ethical committee of Sudan Ministry of Health back then, with it being the only appropriate entity. For the second data collection round, ethical clearance was obtained from Sudan Ministry of Health Research Ethical Committee in February 2021. The clearance was for the overall doctoral study and was clarified in application documents that it is in continuation of the master thesis with follow-up overtime, utilizing the data collected in the first round in 2015. The data for the quantitative sub-study 4 were all secondary data available in public databases and within officially disseminated reports. There was no need for permissions or to seek any ethical approval.

There was no need for any further ethical approval from Swedish authorities as no sensitive personal data was used in this thesis.

Voluntariness and informed consent

Information about the study was shared with targeted respondents. Written informed consent was sought prior to each interview and complemented by oral informed consent at the start of each interview. All contacted individuals were told that they have the right to participate or not, and the right to stop at any time or withdraw without consequences.

Confidentiality, minimizing harm and ensuring safety

The respondents were assured of protecting their confidentiality. While I requested to permit the recording of the interviews, which they all accepted, data (interview recordings and transcripts) were kept securely in password-protected archives, with all the identifying details removed. When presenting the results I tried my best to not include details that can compromise participants' identities.

The collected data focused on the views of entities the respondents represent on EDC commitments, without indulging into any individual personal opinions or political leaning. Yet I recognize the views given might still entail political risk to the respondents and their entities since they included reflections on positions taken regarding specific political changes or governance situation. Given the political sensitivity within Sudan context, I carried discussions with journals we approached to publish the relevant scientific papers to not publicly avail the data. Instead, all data are secured as per Umea University archiving policy and can only be accessed through presenting a reasonable request for scientific purposes, and then only de-identified data will be availed. Respondents were assured of their interviews data safety, and the choice of their work setting as interview location was to improve their feeling of security.

Power dynamics, positionality, sensitivity and respect

My role within both data collection rounds was of a junior researcher, and later a colleague, without presenting any hierarchical pressure. The work relations developed over time and the concerns my positionality bring are presented in the discussion section within methodological considerations.

Results

This section presents an integrated summary of findings from the four sub-studies examining EDC in FCAS. Sudan was a case study for exploring how EDC efforts, challenges, and opportunities are practiced in reality towards enhancing health systems governance. Further a cross-country comparative study investigated if, and how, EDC influenced the progress toward UHC in 17 FCAS. The results are presented within four themes that reflect each on the four sub-studies consecutively, with some additions of relevant cross-cutting insights clarified within the theme when beyond the respective sub-study.

1. The shifting landscape of health partnerships in Sudan: from rigidity to fluidity

This theme depicts the dynamic evolution of health partnerships in Sudan drawing from the first sub-study, with Sudan as the examined case. It reflects on the underlying dynamics of these evolving relations and the interaction of this with the partners perspectives and adherence to EDC. The partners relations altered from a rigid, government-dominated model to more inclusive but fluid - and sometimes fragmented - arrangements. The theme presents findings of **navigating political volatility and power asymmetries** in the health sector partnerships, highlighting how political transitions, donor skepticism, and the contested ownership shaped the evolution of partnerships.

Prior to 2019, Sudan's health partnerships were characterized by a rigid, government-dominated model, where all details of who work where and with who were dictated by the government. Donors were skeptical of the government's legitimacy and hence contested the calls for respecting the government ownership and disagreed with how it was perceived. This was also linked to concerns over priorities representation of actual needs. The government calls for ownership was believed to be a tendency of partners' and fund control under sovereignty claims.

> *Ownership was perceived in a weird way by the Government. {Ownership calling} was a sort of way to grasp money… they were talking about ownership in terms of 'give us the money' … and this is thought of as part of the national sovereignty, dictating everything the partners do and how they do it.*
> (Donor, 2015)

Some donors expressed inability to engage and align directly with the government, given its record on human rights violations leading to subsequent bad international relations preventing it from being part of specific crucial collaboration agreements.

> *Sudan as state has limits due to not signing Cotonou agreement. This creates a difficulty for {us} to be fully aligned within Sudan as development financing for us goes this way. {We} can only consider the needs if they signed this {cooperation} agreement … The Government will not sign the agreement due to legal complications with the ICC and political consequences of its actions and global relations.* (Donor, 2015)

Yet on implementation grounds partner seemed to have compelled by the government directions that they viewed as enforcement, where ownership was often misinterpreted as control. The Government constantly raised the sovereignty flag to enforce controlling ownership claims.

> *If the Ministry wants a health centre building that is pink, we do it pink!* (Donor, 2015)

The national civil society organizations (CSOs) during this era had a limited involvement in the development and implementation of strategies due to marginalization by both the Government and the global partners. They were viewed as implementers not true partners to be fully engaged. Even when involved in some processes, they viewed it as fulfilment of requirements for grants or international pressure.

> *As civil society, …the government doesn't give us that space. The only space they give us is because they are enforced by the donors that they have to involve national civil society.* (CSO representative, 2015)

The civil space was repeatedly misused and intervened in through the creation of pseudo-CSOs that were aligned with the governing regime, Governmental groups under non-governmental disguise as exhibited by one CSOs respondent.

> *The government used to make their own organizations—we call them Government NGOs, GNGOs [laughs]…there was conflict on how to choose the CSOs to participate in the global fund implementation, for example. In the end, people decided… to formulate a network {that unites} organizations genuinely working on health… The {resulting} network is dominated by GNGOs [laughs].* (CSO representative, 2015)

Yet some confined progress on EDCs was witnessed during this period in terms of developing joint health sector strategy and plan and conduction of joint annual reviews that brough together majority of health partners. This progress was attributed to the introduction of good coordination mechanisms such as the Health Sector Partners Forum (HSPF) and the National Public Health Council. However, due to political changes following the revolution (2019), these structures' functionality was not sustained, similar to the other processes of planning and review.

Following the 2019 revolution, Sudan entered a transitional period marked by openness and increased donors' alignment. Coordination platforms such as the HSPF were revitalized, and CSOs gained more respect. Many partners were open to providing support via modalities that were not availed earlier, given the positively viewed changes in the political sphere and the newly appreciated transparency.

> *Since 2019, the scene was getting wider because … the international community viewed {the transitional} government as an approachable entity, and this allowed us to speak loudly… Before, we couldn't. Now we are considering starting a sector budget support.* (Donor, 2021)

However, instability at the sector leadership during this era with weakened technical capacity constrained gaining the fruits of this era's promising partnership scene.

> *The frequent changes at leadership level have caused delays in the strategy release and caused multiple modifications to the priorities set.* (Government respondent, 2021)

> *Often there is the element of turnover … they do not succeed in reaching the full capacity to facilitate the relationship.* (Donor, 2021)

The 2021 coup led to breakdown in partnerships and uncertainty in the overall scene. Donors withdrew and many refused to deal with the coup's regime at all in-order not to legitimize it.

> *After the coup … a lot of support has been frozen from different entities, … World Bank, European Union, USAID, and many others. Many entities and donors are reluctant to deal with the government… many clearly stated that they are not able to deal directly with the coup government.* (Government respondent, 2022)

> *I think the coup is creating an environment of uncertainty … Also, part of the issue is political, related to a high-level decision not to legitimize this coup regime*. (Donor, 2022)

While some collaborations continued, for those remaining projects, the CSOs became the alternative implementers after being at the partnership margins previously.

> *What I'm seeing now is that the donors and some of the international NGOs who were working with the government before started to look at local NGOs as a replacement to do the work, because now there's nothing going with the government*. (CSO respondent, 2022)

These findings highlight how political transitions shaped the partners relations, with partners behavior responding to perceived legitimacy and government stands. CSOs emerged as central actors when the state legitimacy was questioned, and global actors decided to bypass it. coordination mechanisms, which assisted some progress on EDC operationalization, proved fragile under political instability.

2. Navigating complexity and ambiguity: adopting the humanitarian–development nexus in Sudan

This theme addresses the translation process of the HDN approach into practice in a volatile fragile state, and the challenges facing this process due to the existing complexities within this setting. The theme reflects the findings from the second sub-study, supplemented by relevant reflections on the adoption process for the EDC from the overall Sudan's case study. The HDN adoption processes in Sudan was complicated by conceptual ambiguity and disagreement, political turmoil, fragile coordination mechanisms, and tensions among the partners (global and local partners, as well as humanitarian and development actors), similar to that of EDC.

HDN was initially introduced through a top-down approach driven by global actors' interest yet was positively embraced by national stakeholders. The explanations presented by the different partners for this included the commonly agreed need for such approach given the protracted emergency context and the short-term financing of humanitarian projects, as well as the need to implement approaches that derive all partners to invest in local capacities and the system.

> *The nexus is part of the strategic directions for the Federal Ministry of Health … it would support the advancement of the health system in the context of protracted emergencies. The*

> *investment in emergency response is very high, yet we keep having these emergencies because there is no sustainability embedded.* (Government Respondent, 2022)

The Sudanese government introduced EDC agreements (Sudan health compact) for similar reasons. After South Sudan separation, it was obvious that sectors like health were in dire need for effective cooperation to maximize use of scarce resources and hence embraced these agreements.

> *Sudan was a signatory for the IHP+ compact and all relevant global cooperation initiatives after South Sudan separation. That …provided …ground to roll out new approaches that kept coming. So, with those participations in global initiatives, the nexus uptake was much easier for us, in terms of bringing up such approach to Sudan.* (UN agency Respondent, 2022)

Coordination platforms such as the HSPF played a key role in fostering the adoption of the HDN, as they did in advancing EDC implementation.

> *At the HSPF, as partners working on both humanitarian and development spheres, together we discussed how to learn from each other and spread the nexus thinking … The forum approved the {health} policy, so the discussion transformed into operationalization.* (UN agency Respondent, 2022)

However, despite these efforts, conceptual ambiguity for the HDN understanding persisted. Each actor took the concept into a direction based on their interventions' nature and interest.

> *There is no one and only nexus concept in Sudan … Darfur development strategy revival that started in 2013 argued moving from humanitarian phase toward development phase, then the European Commission was operationalizing the systems resilience as nexus, … the United Nations … was defining the so-called collective outcomes, and also the World Bank with the cash transfer claiming to move away from humanitarian work.* (Donor, 2022)

On top of the ambiguity, tensions between humanitarian and development actors persisted. Development partners blamed humanitarians for not being serious about the nexus and embraced it only as a way to gain access to the increasing long-term development financing, especially during the transitional period. It was even seen by some development partners that humanitarian and development principles

were not compatible, where alignment and mutual accountability don't match with core humanitarian principles like independence.

> *humanitarians as they see a crisis in countries like Sudan, they don't have the money to continue financing but want the development partners to take over funding the kind of work that they have been doing … Humanitarian and development principles are different and mutually exclusive … you cannot be independent and aligned at the same time* (Donor, 2022)

Humanitarian actors, on the other hand, were new to collaborating with national authorities as HDN and EDC required. They reported experiencing inefficiencies in the national systems they couldn't deal with. The humanitarians claimed that strengthening the systems was the development partners' responsibility that, despite prolonged investment, couldn't achieve the requirements, so how were they expected to manage such tasks beyond their mandates of saving lives and alleviating suffering.

> *As humanitarians, our duty continues to save lives and for that, our interventions continue despite any politics. But to apply nexus for sustainability, we need to wait for the other side to start building towards our way as well so that we can meet somewhere in the middle.* (Humanitarian respondent, 2022)

Additionally, the political instability, especially after 2021 coup, undermined the big ambitions to advance the HDN implementation.

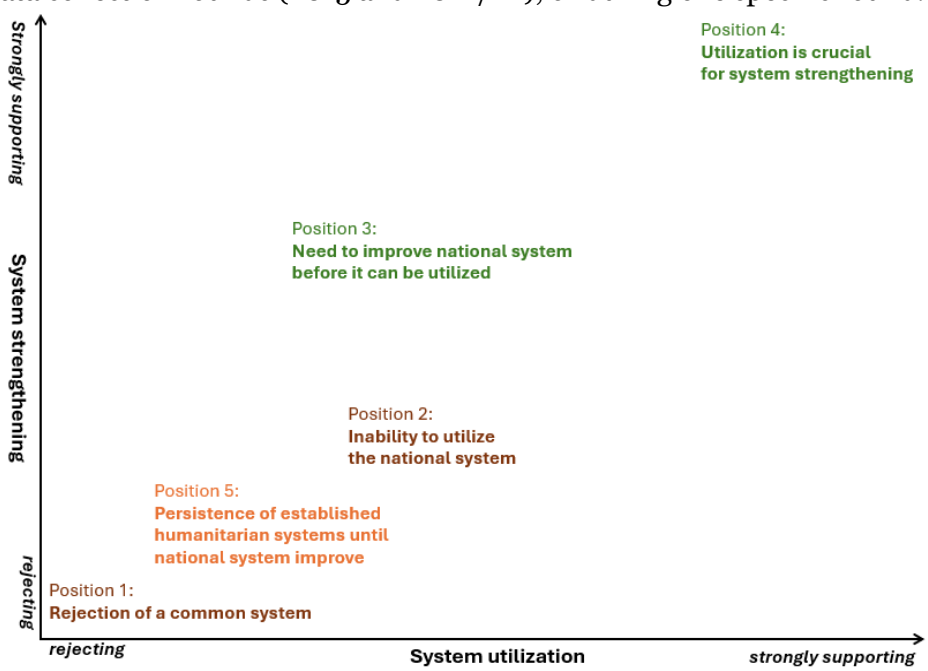
> *After October 25th {2021 coup} … development donors suspended their support… Humanitarian actors exist, and government exists -with low capacity-, but development partners had disappeared. So, we have a broken triangle.* (Humanitarian respondent, 2022)

The ambiguity, instability, disagreements and tension that undermined the HDN adoption is comparable to disagreements and tensions recorded on EDC adoption among various partners levels during the different eras. The contested meaning of ownership from the first theme is a clear example of such contestations.

3. Between parallelism and integration: Utilization vs strengthening of national health system components

This theme highlights the tension between maintaining parallel systems and integrating in the national health system from the third sub-study. Debating strengthening versus utilizing the national health system, it reflects dilemmas of the humanitarian-development divide and also motives shaped by the broader political-partnership dynamics changes.

Significant position(s) changes were seen between 2015 and 2021/22 on strengthening and utilizing unified national health system components, particularly for procurement, financial management and information. Figure 9 below outlines the major positions found among the partners at these two timelines. Stands on system utilization are depicted on the horizontal x-axis ranging from rejection at the origin to strong support on the right end. System strengthening attitudes are represented on the vertical Y-axis ranging from rejection (origin) to strongly support (top). The coloured legends explain if the specific position is found within both data collection rounds (2015 and 2021/22), or during one specific round.



Brown: position from 2015. *Orange: position from 2021/22.*
Green: position from both rounds.

Figure 9. Partners positions' map on system strengthening vs utilization

In 2015 various partners (INGOs, some donors and UN) took a stand of **rejecting the notion of a common health system**. They argued that this is not practical, and instead it will cause inefficiency. It is seen as more bureaucratic layers that would add unnecessary duplication of processes, as for the remote places where NGOs provide services the Government cannot access or lead on anything there anyway.

> *For me, there is no reason to have unified system. We discussed {basket financing} for some time but did not reach anywhere. I don't think it will be feasible now, because why do we add layers? This way things go to the government and from the government to the NGOs, national or international, as they are the ones who implement in many areas. Now things go to NGOs so why do we need to put a government layer?* (INGO respondent, 2015)

To them, the national financial management system particularly was weak, and many couldn't invest on improving it as it was beyond their mandate or that they were too small to invest on such big work. They further reported a history of bad experiences with implementation through government authorities including claims of corruption, misuse, low capacities and delays in reporting leading to losing and returning funds that are needed.

> *Our preference is not to go through any government system or public sector, especially for funding, mainly because reporting requirements are not met. There is suspicion of funds misuse, corruption. We are small compared to other UN agencies, and our funds are much smaller. We don't feel it will make a difference. Others are working in health system strengthening, financial management system strengthening, procurement and supply chain management, but we are not"* (UN agency Respondent, 2015)

These partners argued similarly that the procurement and supply system component suffered issues of mistrust and capacity gaps, claiming their own rules and systems would bring more efficiency and ensure quality.

> *We don't follow this common system idea. Right now, we have our own procurement rules. We know the Sudanese procurement rules, but {our} government put rules for us that we need to compare and check also private companies and compare the prices and the quality to find the best choices. We follow these rules.* (Donor, 2015)

Inability to utilize the national system was expressed in 2015 mostly by CSOs and a donor, denoted for different reasons. They didn't strongly reject the common system's notion, but the CSOs indicated a problem with the government not allowing them to utilize components they feel they could benefit from using. This contrasts the Government active take to bring other partners - particularly international - to use it (depicted in the fourth position).

> *As civil society we can't use the national system because the government don't give that space. The Government don't enable us to use their procurement and supplies system like they are trying to with big donors and UN agencies.* (CSO, 2015)

A donor argued that political consequences of not signing agreements due to the ICC warrants and subsequent sanctions blocked them from utilizing the national system.

> *We cannot use the country system because Sudan did not sign the Cotonou agreement. That is why the rules prevent us from using the national system.* (Donor, 2015)

These two positions were no longer seen during 2021/22 and that could be largely explained by changes in the political atmosphere, and the shifts in relations between the post-revolution authorities and other partners.

Two positions persisted within both timelines (2015 and 2021/22). These reflected a dialogue between the **need to improve national system before it can be used**, and on the other side how **utilization is crucial for system strengthenin**g. The later viewed utilization as pre-requisite for any strengthening efforts in contrast to the first putting strengthening as a condition to utilization.

Those who took the stand of the need to improve the national system before it can be used were dominantly UN agencies, Development Banks and some donors by 2015, and it was taken by a donor, INGO and one Government representative by 2021/22. This position takes an active step from outlining the system weaknesses into acting on it.

> *Sudan system lacks transparency and accountability. There are issues related to the budget adoption and the distribution of funds to different sectors including the security sector, these are the issues that as donors we look at. Parts of our programs and funds were to provide technical assistance as we support*

> *the Ministry of Health to strengthen the system and improve the public financial management (PFM) system.* (Donor, 2015)

The position outlining cruciality of utilizing the system while simultaneously working on enhancing it was the one with the utmost strongly supportive stand for both aspects of strengthening and utilizing.

> *When we talk about the system, improving is by utilizing it. We can't put it in shelves and assess and design and later say look it is now beautiful come take and use it.* (Government, 2015)

In 2015, this stand was taken solely by the Government, while in 2021/22 it was shared with donors, UN agencies and even one INGO. This could be due to the above discussed openness in collaborating with the Government seen during the transitional period, which further explains partners' movement from the stands of rejecting utilization or demanding a strengthened system first in 2015, into engaged stands of actively working on strengthening the system and concurrently utilizing it.

The UN agencies' collective progress into utilizing the system can be further attributed to their cumulative piloting experience, assessments and strengthening efforts.

> *Since 2014 we have been trying, with the government, to pilot this "HACT" approach … that is supposed to be the approach for the agencies of UNDP, WFP, UNFPA and UNICEF… with this HACT the idea is to trust the national financial management system, they don't have to send us all the documents. The Government keeps the receipts, but occasionally we will come to do a spot check.* (UN agency a, 2015)

> *We cannot rely on kits procured through humanitarian operations. We agreed these should be merged into the national system and we started few years ago. Further for the information we also said all sources should fall into the District Health Information System 2 (DHIS 2). We are now working on including surveillance components that were managed separately into the DHIS2.* (UN agency b, 2021)

The introduction of the HDN approach in Sudan contributed to the shifts of most INGOs from rejection of utilization in 2015 into more actively working on strengthening and piloting the system use, to even some engaged in utilizing and strengthening simultaneously.

> *as we are piloting this nexus approach to build the capacity of the state and locality level authorities, we had to start supporting the national system so that we can be able to use it in the future.* (INGO respondent, 2021)

Interestingly there was a government respondent who actually supported the stand of strengthening as a pre-requisite for utilization by 2021/22, entailing that the Government was no longer taking collectively the position of strongly pushing towards further utilization as in 2015. This could be explained by the political changes from a central controlling government to the broad-alliance transitional Government and shed light on lack of unified vision during the transition outlined in the first theme.

The resistance for system utilization witnessed by 2021/22 was only seen among humanitarian actors claiming the **persistence of established humanitarian systems until national system improves**. They believed they had better functioning systems than the national system despite years of investment in the later. They attribute the capacity gaps to political instability and economic constraints. Within their position they had a passive view of their role, putting the responsibility on others (the Government and development partners).

> *The national system is weak, even after years of external support, due to political and economic issues… If we crossmatch what our emergency partners systems do, and the national system, they aim to give same results. As long as my system can function better than the national system then I will keep using them.* (Humanitarian, UN agency, 2022)

> *Each of the National Organizations has their own system. We need a well-established national system then the Government can tell us to abolish our systems to utilize their one system.* (CSO, 2021)

The national CSOs by 2021/22 no longer complained of being prevented from utilizing the system, as in 2015, but their stand (as most humanitarian actors) persisted on utilizing their own systems.

The rejection of, prevention from or resistance to utilization were often focused on the financial management system and to a lesser extent on the procurement and supplies system, while all partners showed interest in having a common health information system, with various degrees of support to its strengthening efforts.

4. EDC in the balance: alignment, joint technical assistance and the pursuit of UHC in FCAS

This theme zooms out from the case study findings that examined the specific real-life EDC implementation into examining the effect of EDC principles on attaining health targets in wider FCAS. It presents mainly the findings of the cross-country comparative study (sub-study 4). Hence, it puts the relation of EDC and health system governance in the balance of contribution toward achieving health goal. The comparative study design allows for reflecting on it through what we learned from Sudan case study.

The cross-country analysis looked into the 17 examined FCAS and if their implementation affected the attainment of key health targets, focusing on Universal Health Coverage (UHC) and Under Five Mortality Rate Relative Change (U5MRC). The results reflected which of the examined six EDC principles were necessary or sufficient conditions to attain the set targets. Table 2 below shows the results of the necessity and sufficiency analysis.

Table 2. Analysis of necessity and sufficiency toward UHC

Condition	**Necessity analysis**		**Sufficiency analysis**	
	Consistency	***Coverage***	***Consistency***	***Coverage***
Alignment (to priorities)	0.273	1.000	1.000	0.273
Partners coordinate	0.182	1.000	1.000	0.182
National coordination mechanism	0.182	0.667	0.667	0.182
Budget alignment	0.182	1.000	1.000	0.182
National Monitoring and Evaluation (M&E)	0.182	0.667	0.667	0.182
Joint Technical Assistance (TA) Plan	0.182	1.000	1.000	0.182

No single condition was found to be necessary to attain UHC index of 40 or above (no single condition met the necessity consistency cut off of 1). Four conditions (alignment with national priorities, partners coordination among each other, financial support aligned with national budget, and having a joint technical assistance plan to mainstream technical support) were sufficient (consistency=1, above threshold of .75) to achieve the UHC threshold, meaning that whenever any of these conditions was present, UHC (40 or above) was also present. The coverage for the sufficiency analysis was low; indicating 27% of the FCAS that achieved the outcome can be explained by the presence of alignment, and 18% by the presence of each of the conditions of partners coordinate, budget and join TA respectively. These coverage percentages indicate that there were few countries with the conditions in comparison to the overall countries achieving UHC index of 40 or above. Similar results indicating these same four sufficient conditions were found within the analysis toward U5MRC with a slightly higher coverage for these exact sufficient EDC principles (see in Annex 4).

Further a parsimonious solution model was produced to describe the possible pathways these conditions had toward the set targets. Table 3 below depicts the parsimonious solution.

Table 3. Parsimonious solution model towards UHC

Solution model: Align + Joint TA → UHC

	Consistency	**Raw coverage**	**Unique coverage**	**Covered cases**
Align	1.000	0.273	0.273	Comoros, DRC, Eritrea
Joint TA	1.000	0.182	0.182	Mali and Zimbabwe
Model	1.000	0.455		

Above solution pathway consisted of two conditions with OR (+) relation: (1) the existence of good alignment, or (2) joint technical assistance (TA), either of which was sufficient alone to attain the positive outcome of UHC.

The model covered 45% of cases with UHC over 40 (total raw coverage). Alignment alone explained 27% of the cases with UHC index over 40 (unique coverage of 27%), including three countries within the set: Comoros, DRC and Eritrea. Joint TA explained 18% (unique coverage) of cases with UHC index above 40 (Mali and Zimbabwe).

The parsimonious solution towards U5MRC was similar to the above model for UHC, and the cases falling within the model pathway to U5MRC were identical to those within the UHC model corresponding to each exact sufficient condition (see in Annex 4).

These results show that within our dataset of 17 aid recipient FCAS, alignment with national priorities and technical assistance were central in achieving health targets of UHC and U5MRC. Although these principles were not necessarily present in every case that achieved the health targets, their presence covered a moderate proportion of the cases. This goes in line with the theory of change presented in this thesis and highlights that pathways from EDC to health and development targets are multiple and can be more unique in the complex setup of FCAS. Further within this sub-study we saw that adherence to a principle like alignment was inconsistent among these 17 FCAS, just as in the Sudan case, often driven by factors like legitimacy and regime perception when we examined these cases further in the literature.

The centrality of alignment and Joint TA emphasizes the importance of focusing on national capacities and system strengthening while limiting fragmentation of support that can hinder the effort toward establishing a unified efficient national system(s). This puts emphasis on the governance functions of engaging in policy and strategy formulation and enhancing implementation levers through effective partnerships and collaborations.

Discussion

The findings across the four sub-studies reveal a complex, interdependent relationship between EDC principles and health systems governance in FCAS. The Sudan case illustrates how political context, coordination effort and partners behavior interact to shape the real way EDC is operational. The comparative analysis in sub-study 4 reinforces these insights by showing that alignment and joint technical assistance are not theoretical ideals but practical enablers of progress on health system goals.

Alignment, joint technical assistance and coordination can influence health systems and development targets, but …

Findings across the four sub-studies revealed a reinforcing feedback loop (Figure 10) that elaborate how EDC principles were translated and interacted in FCAS. Initial progress notably begun with instituting ***coordination platforms*** such as Sudan Health Sector Partners Forum that enabled partners dialogue on EDC commitments. The activation of these mechanisms created the procedural space for partners to advance ***alignment*** through jointly assessed and developed policies and strategies, and further to plan ***joint technical assistance*** targeting critical governance levers—public financial management, procurement, and information system components. Joint TA then enabled incremental ***system strengthening***, which improved trust among partners to ***utilize the national system*** rather than maintain parallel structures. Utilization itself accelerated further strengthening by exposing gaps and driving iterative improvements, thereby reinforcing trust and partnership dynamics, especially during periods of progression on partnership. This confidence made deeper alignment operationally feasible, sustaining coordination and promising more ambitious reforms, under the condition of enabling political atmosphere. The comparative analysis confirmed that alignment or joint TA alone were sufficient conditions for achieving UHC and the other development goals in several FCAS, underscoring that multiple entry points into this loop can yield progress. The findings show that alignment, joint technical assistance, system strengthening and system utilization do not operate as isolated interventions; they form a reinforcing process that—if maintained—can transform fragmented aid into resilient system and yield major health result. This was notably possible within specific political window of opportunity, and hence the use of the "but" in the heading refers to conditionality within political context.

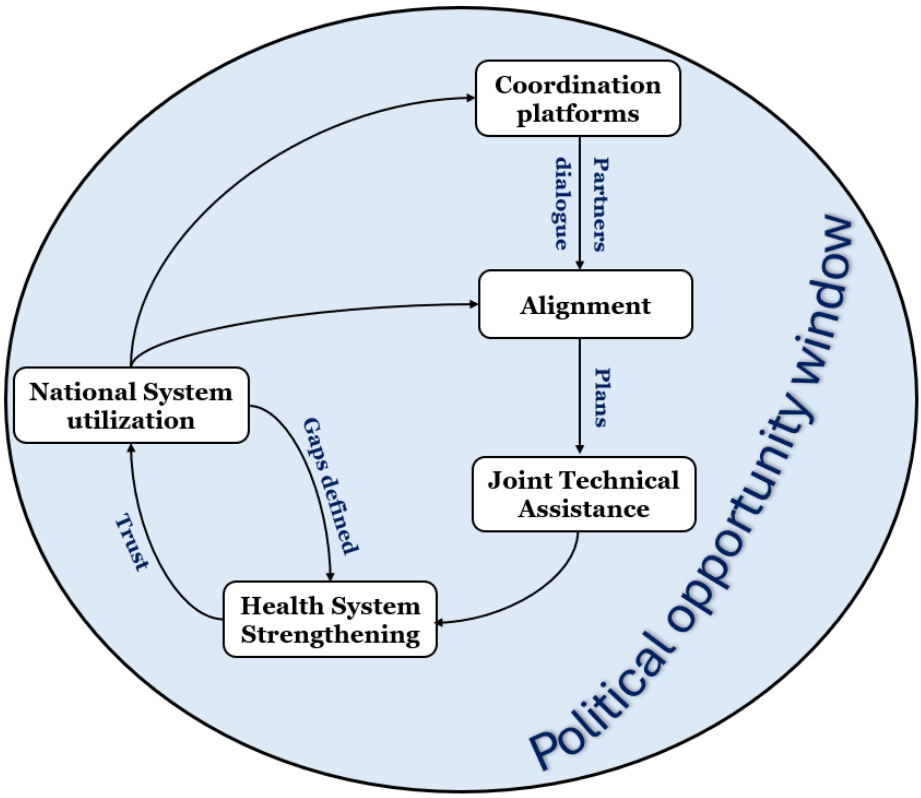


Figure 10. Reinforcing feedback loop of EDC and health system in FCAS

Coordination platforms as the conversion mechanism from principles to practice

The Sudan's Health Sector Partners Forum (HSPF) and the National Public Health Council functioned as conversion mechanisms—the spaces where alignment and TA intentions became joint strategies, endorsed policies, and co-managed roll-outs. The HSPF enabled cross-constituency debate (government, donors, UN, INGOs, CSOs) on using single information and supply systems and on applying the HDN approach in protracted emergencies context. When these forums functioned, partners coordinated better among themselves and with government, and concrete system-use pilots followed. When these forums were weakened by political shocks, fragmentation returned quickly.

The comparative study affirmed this picture: partners' coordination was associated with target attainment as "partners coordinate" was another sufficient condition for improving UHC and other development targets

(though with lower coverage than alignment and joint technical assistance and was not part of the final solution model). This further highlight that coordination is not merely procedural—it is outcome-relevant in FCAS.

Decades of aid-effectiveness work at both the humanitarian and development domains anticipated that coordination reduces duplication, clarifies roles, and enables managing for results (Akl et al., 2015; Alkhalil et al., 2024; McKee et al., 2020). Health-sector partnership compacts, operationalization of joint reviews, inclusive strategies development, and mutual accountability arrangements can all be enhanced through coordination platforms (Carneiro, 2024; OECD, 2005; WHO, 2013). These were the exact instruments Sudan attempted to advance through the HSPF. Studies on relational coordination show that when actors agree on goals, roles, and communication systems, they achieve faster, more accurate, and mutually respectful coordination that enables managing for results (Bolton et al., 2021). Hence coordination is a cornerstone for EDC.

Alignment as a catalyst for system-first practice and results

In the Sudan case, there was a general agreement on the alignment importance, despite contestations over ownership during periods of problematic government legitimacy. The period of improved political openness (2019-2021) witnessed critical advancement on alignment to nationally agreed strategies and plans, including donors considering sector budget support. It further led to augment stands on system strengthening importance and utilization of unified system arrangements by even some humanitarians. Conversely, political shocks (the 2021 coup) disrupted alignment and reversed gains, pushing partners toward parallel channels. These dynamics show that alignment is feasible in FCAS but is context-sensitive. Progress on alignment exacerbates during political opportunity window to allow partnership reforms.

The comparative analysis (sub-study 4) went further to show how alignment with national priorities was a sufficient condition for attaining an essential UHC threshold (UHC ≥ 40) and other health development targets. In the parsimonious model, alignment alone explained progress in several FCAS cases (e.g., Comoros, DRC, Eritrea). While not necessary in every instance, its sufficiency underscores that when alignment is present, health target attainment follows in a meaningful share of FCAS.

The findings also support the EDC intended shift from donor-driven programming to country ownership and alignment (Deutscher & Fyson, 2008). Further, they align with the IHP+ "seven behaviours" (UHC2030, 2013) that operationalize alignment through using and strengthening national PFM, procurement, and information systems -rather than bypassing them. Global literature indicates that when partners align, priorities are addressed (McKee et al., 2020), incentives to use national systems rise (Health Data collaborative & UNICEF, 2022), efficiency improves (Paul & Vandeninden, 2012) and mutual accountability becomes thinkable (Vince et al., 2024).

Joint technical assistance as the bridge between intent and system capacity

Sudan's case emphasized that how technical assistance is organized matters. Jointly planned TA—anchored in a single sector strategy and nationally lead arrangements— helped move many partners from "strengthen-before-use" stalemates toward "use-to-strengthen" trajectories. Joint piloting of the HACT approach was reported by UN agencies towards reliance on the Public Financial Management system component, merging supply lines into the national procurement and distribution system, and consolidating surveillance streams into DHIS2. These steps were all anchored on joint technical assistance, and increased confidence in these national system components, unlocking further system utilization (if not for negative political shifts).

In sub-study 4, joint technical assistance plan was the other sufficient condition for UHC (distinct from alignment), explaining additional positive cases (e.g., Mali, Zimbabwe). This indicates that, even where full alignment is politically or administratively out of reach, coordinated TA can still tip the system toward results by targeting bottlenecks in governance functions.

Within the SDG3 Global Action Plan and IHP+ agenda, "well-coordinated technical assistance" is a core behavior (WHO, 2019). This is because fragmented, partner-specific technical assistance often overloads recipient countries ministries, duplicates tools, and leaves capabilities unrooted in national routines. Joint TA reorients incentives toward shared products (guidelines, platforms, data standards) and shared capacities (PFM, procurement, M&E), making subsequent system

utilization lower-risk for all actors, including humanitarian responders seeking reliability during crises (Solter & Solter, 2013; Ward et al., 2024).

FCAS are volatility-prone, hence relying on one "big lever" rarely works. Our findings suggest equifinality, i.e. multiple paths toward an outcome (Mello & Ostermann, 2022). Alignment can unlock outcomes even when TA or coordination is imperfect; joint TA can deliver results (UHC) when full alignment is constrained; and functioning coordination fora can reignite stalled processes aftershocks. Although in sub-study 4 no single condition was necessary, yet a couple were individually sufficient. This permits the courageous proposition that EDC may allow multiple entries for health systems governance enhancement and progress on health goals.

Political context and transitions as anchors for EDC adherence

The Sudan case illustrates how political context operated as an anchor for adherence to the EDC principles. Periods of political openness—such as the post-2019 revolution— improved alignment of partner support with national priorities, revitalized coordination fora, and encouraged discussions on sector budget support. Conversely, instability, exemplified by the 2021 coup, reversed these gains almost immediately, freezing donor funds and dismantling joint planning processes. This volatility underscores that EDC commitments are not self-executing; they are contingent on the broader political environment.

Before 2019, alignment was periodically constrained by donors' concerns over regime legitimacy and international legal restrictions, including Sudan inability to sign agreements like Cotonou due to ICC indictments. These constraints limited direct engagement with the government and the national system components, even when formal EDC compacts existed and signed by Sudan and partners both globally and internally. Yet coordination platforms like the Health Sector Partners Forum preserved fragments of cooperation by enabling issue-specific technical assistance and dialogue among actors, demonstrating that partial adherence can persist even under adverse conditions.

This experience challenges the reductionist view that health system problems can be solved through technical fixes alone. Birn et al. (2017c) argue that the dominant belief that health challenges are purely technical ignores their political and economic underpinnings. Addressing these

political determinants requires reconfiguring underlying power dynamics (Brugger et al., 2025). Governance and EDC enhancement, thus, demands both technical and political strategies. Investments in systems must be coupled with efforts to navigate legitimacy dilemmas, sovereignty claims, and donor risk perceptions (Kaseya, 2025; Kilby, 1999).

Legitimacy, sovereignty and health needs: steering a pressing predicament

In FCAS, the tension between addressing urgent health needs and avoiding the legitimization of oppressive regimes is persisting. Voon (2002) notes that this dilemma is rooted in fragility's dimensions—contested legitimacy and recurrent conflict—alongside the weak service capacity. Calls to strengthen local institutions, central to EDC principles, become politically fraught when the state's authority is disputed or associated with human rights violations. Development actors face a paradox: investing in national system risks reinforcing such regimes yet bypassing them perpetuates fragmentation and undermines long-term development goals. Sudan exemplifies this challenge as seen in the above. The reported patterns underscore Carment and Samy (Carment & Samy, 2023) argument that aid does not operate in a vacuum; it is deeply embedded in political realities. Legitimacy dilemmas shape not only the feasibility of alignment and harmonization but also the modalities of technical assistance, forcing actors to balance sovereignty claims against accountability and human rights considerations, and perceptibly political interests of other partners. In fact, in similar illegitimate authoritarian regime cases, as Syria Assad's, cries to respect national sovereignty were seen as a facade, masking systematic obstruction to humanitarian access and essential services (Leenders & Mansour, 2018).

Current war in Sudan exemplifies these dynamics in their starkest form. Although the sub-studies data pre-date the war, it is possible to forecast the partnership trajectory based on what is learnt from our sub-studies and comparing similar context and actions. Partners now face the dilemma of engaging with warring parties while safeguarding humanitarian principles and meanwhile trading with the "legitimacy" questions in the face of sovereignty calls, expectedly. Reports of demands for control over humanitarian operations (Feltman, 2025), attacks on aid convoys (Al Jazeera, 2025), and the recent expulsion of WFP leaders (World Food Program, 2025) echo earlier patterns of sovereignty

contestation and politicization of aid. These behaviours suggest that current cooperation will hinge on adaptive strategies, leveraging vital coordination spaces with cautious involvement with warring parties' governments, empowering civil society as alternative implementers, and designing flexible technical assistance that can operate amid fragmented authority. In such contexts, sustaining even partial EDC adherence becomes a political act as much as a technical one.

The humanitarian–development divide and the strengthen vs utilize dilemma

Patterns seen in FCAS in which humanitarian short-term mandates and postures often clash with long-term governance and system goals is highly present in Sudan's health sector case of implementing EDC and the nexus. In the sub-studies, humanitarian actors tended more to maintain parallel financial, procurement, and information systems, arguing that national system was too weak, slow, or politically compromised to meet life-saving urgent needs. Development partners and government, by contrast, increasingly contended that responsible utilization of the national system is itself a core strengthening strategy—where systems mature through use, not only through "off-line" technical fixes. This divide is well-documented in the global health governance literature. Persistent fragmentation and parallelism endure where multiple actors have different incentives, time horizons, and accountability lines, especially in FCAS (Bogale et al., 2024; Spicer et al., 2020). The findings from Sudan case study align with this as initial resistance to system use was strongest among humanitarian agencies, while the transitional opening after 2019 saw greater willingness by development partners to move, as commented earlier, from "strengthen-before-use" toward "use-to-strengthen".

Deliberations around the humanitarian–development nexus in Sudan—through the Health Sector Partners Forum—helped shift positions for some actors by creating a venue to negotiate sequenced risk management. Partners used these forums to embed joint TA into confidence-building steps: HACT approach with spot checks as risk-based assurance in public financial management; phased integration of procurement and supply chain so that kits and vertical pipelines gradually moved through national procedures; and DHIS2 consolidation to reduce duplicative reporting and bring surveillance streams into a single information backbone, as hoped.

These measures, documented through sub-studies 2 and 3, operationalized the partnership behaviours of "well-coordinated technical assistance" and "use of country systems," turning abstract principles into implementable safeguards that narrow the humanitarian–development gap by lowering perceived risk to service continuity (UHC2030, 2013; WHO, 2014). Hence, joint TA acted as a bridge between the immediate needs' logic of humanitarian action and the institution-building logic of development cooperation as reflected above.

The humanitarian-development divide is found to have a normative aspect, pitting humanitarian principles of independence and neutrality, against alignment, harmonization, and mutual accountability principles. In Sudan, some respondents argued that independence and alignment are "mutually exclusive" in practice—especially under contentious authorities (sub-study 2). The literature similarly highlights this collision of principles where humanitarian doctrines prioritize autonomous access and speed, whereas development cooperation emphasizes country ownership, system use, and mutual accountability (Saez et al., 2021). This study suggests these opposing positions can be partially reconciled when alignment is framed as alignment to "nationally-agreed" priorities developed through multi-stakeholder platforms (rather than to a single political centre), and when assurance measures (e.g., monitoring, audit and data transparency rules) are jointly specified.

If I would question if development partners' stand on alignment is neutral, Sudan's trajectories show it seldom is. Before 2019, several donors' alignment decisions were constrained by legitimacy concerns and international legal restrictions. After the 2021 coup, freezes and re-routing through non-state channels returned (sub-study 1). This reiterates a key theme in the aid effectiveness literature; alignment is politically shaped by perceptions of governance quality, international relations, and domestic accountability, not merely by technocratic strategy documents (Barnes & Brown, 2011; McKee et al., 2020). The system strengthening vs utilization controversy thus cannot be treated as a purely technical sequencing problem. It is a political choice about who is trusted to manage public resources, which forums are legitimate to set priorities, and what risks donors are willing to carry in volatile settings. In this sense, what appears as a technical argument concerning "strengthen vs. utilize" is often humanitarian–development politics by another name.

The practical path forward in FCAS is not to collapse a logic into the other, but to institutionalize negotiated middle paths that keep inclusive coordination forums functioning even during politically stressful times; openly discuss risk-mitigation measures that ensures gradual utilization combined with the application of targeted technical assistance; and agree on common standards (PFM, procurement, HIS) that all actors—humanitarian and development—commit to apply wherever feasible.

Monitoring and accountability gaps

Both Sudan's experience and the comparative analysis in sub-study 4 underscore a critical weakness in the current architecture for tracking EDC in health. Monitoring frameworks remain perception-based, infrequent, and poorly harmonized. In Sudan, despite formal commitments to EDC and specifically to manage partnership for results, since the end of the IHP+ assessment rounds and its shift to UHC2030, there was no systematic mechanism to verify whether partners actually adhered to commitments and hence whether EDC practice progressed or not. Further, there are no measures of shifted behaviours—such as using national systems or coordinating technical assistance—beyond reporting in a story-telling manner in joint reviews. The SDG3 GAP indicators applied across FCAS rely heavily on government perceptions of partners' commitments, rather than measuring observable changes in practices that advance EDC. This creates a disconnect between global accountability narratives and real progress on the ground. Similar gaps on the EDCs reporting are found beyond health. Gisselquist et al. (2024) indicated that shortcomings in the data explain much of the lack of evidence on EDC impact. They went further to suggest that some shortcomings could be easily fixed with adjustments to the indicators and data collection techniques.

The comparative sub-study revealed that even where alignment or joint technical assistance correlated with improved health outcomes, the mediating system changes were not captured by present monitoring tools. Assessments are sporadic, with long gaps between reporting cycles, making it difficult to track incremental improvements or identify early warning signs of regression during political shocks. Moreover, the multiplicity of health partnership platforms as the UHC2030, SDG3 GAP, and the evolving architecture of Global Health Initiatives (GHIs) including the Global Fund, Gavi, the Global Financing Facility (GFF),

along other actors from the emergency domain —as platforms examples— pulls actors in different directions, fragmenting accountability efforts and limiting cross-learning.

The coexistence of multiple global health partnership platforms creates a highly polycentric governance environment. While these platforms share overarching goals of accelerating progress toward UHC and SDG3, their distinct mandates, governance structures, and operational modalities often pull stakeholders in different directions. This multiplicity can fragment accountability mechanisms and limit opportunities for systematic cross-learning. In fact, while the health sector started EDC commitments monitoring early on through the IHP+ (currently UHC2030), the multiplicity of collaboration platforms currently can be seen as a setback.

The fragmentation and overlap within mandates and mission is striking. For example, UHC2030 primarily focuses on health systems strengthening and promoting alignment with national strategies, with principles of equity and financial protection emphasis (UHC2030, 2016a). In contrast, SDG3 GAP seeks to foster collective action among 13 major global health agencies to accelerate progress on health-related SDGs (WHO, 2019). Meanwhile, GHIs such as the Global Fund and Gavi continue to prioritize disease-specific interventions—HIV, TB, malaria, and immunization—while the GFF targets financing for reproductive, maternal, newborn, child, and adolescent health. Emerging actors like CEPI (focused on epidemic preparedness) and Unitaid (market-shaping for innovative health products) introduce additional layers of complexity.

Empirical evidence from FCAS underscores the governance challenges posed by this multiplicity. In South Sudan, humanitarian health clusters coexist with development-oriented health sector working groups, creating parallel planning and reporting structures (Qaddour et al., 2025). Similarly, in Afghanistan, donor-driven procurement established systems undermined national ownership and accountability. These fragmented arrangements increase transaction costs for ministries of health, dilute responsibility across multiple actors, and hinder the harmonization of priorities—contrary to the aid effectiveness principles of alignment and harmonization (Basij-Rasikh et al., 2024; Paterson et al., 2025).

Without adjustments to the way these global partnership platforms' function, fragmentation will deepen within FCAS hindering results.

Further, monitoring will continue to lag behind reality, and accountability will remain symbolic rather than substantive, particularly in FCAS where political volatility demands agile correction. If left unreformed, these gaps risk undermining progress toward SDG3 in FCAS, as weak accountability erodes trust, perpetuates fragmentation, and delays the systemic changes needed to achieve universal health coverage and other health targets.

Methodological considerations

This thesis employed a mixed-methods design integrating three qualitative studies conducted in Sudan and a quantitative cross-country comparative analysis. This section synthesizes reflections on trustworthiness and rigor, as well as the limitations.

Trustworthiness for the qualitative sub-studies (1, 2 and 3) was improved through the application of several measures guided by Dahlgren et al. (2019, pp. 42–50) and related guide of the method within each sub-study. Semi-structured interviews were conducted in two rounds (2015 and 2021/22). Respondents represented government, donors, UN agencies, development banks, INGOs and CSOs across both humanitarian and development constituencies (total 31 interviews). This diversity of respondents' voices contributed substantially to the findings' **credibility**. Triangulation was further enhanced by comparing interview insights with documents such as partnership agreements, reports from joint reviews and notes from coordination meetings, several assessments of financial management, health information, and procurement systems, and policies. Credibility was strengthened by iterative adaptation of interview guides in all qualitative sub-studies, allowing exploration of emergent issues such as truthfulness of statements like adhering to national policies when did not even read it (sub-study 1), HDN conceptual ambiguity (sub-study 2) and shifting attitudes toward national system utilisation (Sub-study 3). The different analytical approaches utilized in the three sub-studies each provided an analytical structure grounded in the data and suitable for their respective research questions. Framework analysis supported comparison within each respondent's input and between the respondents, as well as over the timeline, which was necessary for sub-study 1. The choice of the policy triangle to assist organizing themes from sub-study 2 allowed easy understanding of the complex issue of the HDP nexus. Opting for situational analysis with positional maps in sub-study 3 enabled visualization of partners positions even with changes over time.

The insider–outsider initial coding approach in sub-study 1, combined with extensive memos and cross-checking of interpretations, helped mitigate bias and anchored interpretations in respondents' narratives. This process was replicated with the co-authors (supervisors) for the second round of sub-study 1 and the other two sub-studies.

Dependability was ensured through detailed documentation of methodological decisions, including coding matrices, decision trails, analytic memos, and reflexive notes. Examples of these are shared within respective papers. Comparison across two political shifts for three eras — pre-revolution, transitional, and post-coup phases— further enabled consistency checks by revisiting previously identified patterns and observing how they shifted in response to context change (sub-studies 1 and 3).

Confirmability was strengthened by the involvement of co-authors and a colleague without prior engagement in Sudan's health system, who provided external analytical perspectives. Transparent coding process was ensured. Their participation in interpreting themes and reviewing positional maps reduced the influence of the primary researcher's insider knowledge and professional history within the sector, with all decisions documented. The above explained triangulation scope also assists the confirmability. Further, the use of quotes and memos in reporting links data and findings. Reflexivity was explicitly maintained throughout, acknowledging how my personal proximity to the health sector and political events may shape interpretation. It is reflected on further below.

Transferability was supported by presenting detailed descriptions of Sudan's context, political transitions, health sector architecture, coordination mechanisms, and the evolution of partners' (humanitarian and development) engagement. These contextual accounts allow readers to assess how insights could apply to other fragile settings where donor fragmentation, governance instability, and parallel humanitarian systems are common.

These three sub-studies focused on insights from health sector actors. Having actors of non-health origin could have provided valuable insights into the topic. Their perspectives could have highlighted different challenges, opportunities, and ways to improve coordination, building on the health sector experience where some are engaged through the HSPF (e.g., Ministries of Finance, social welfare, foreign affairs, etc.).

Addressing the comparative nature of sub-study 4, QCA allowed the examination of this important relation between EDC and health goals, with consideration of various solutions for the outcome and its absence, in the presence and absence of the conditions. The supplementary analysis conducted confirmed the results and ensured the rigourness. Given the limited number of comparable FCAS globally, data availability limitation, and the complex, non-linear pathways linking EDC to health outcomes, QCA offered a suitable methodological approach.

Rigourness of sub-study 4 was further sought by relevant measures. **Internal validity** was supported through transparent calibration of the EDC conditions and UHC index using theoretically motivated thresholds.

Reliability derived from reliance on official secondary data sources (WHO and World Bank, and the UN SDG database) and replicable analytic procedures performed using freely available R packages for QCA. The stepwise analytic process—necessity analysis, sufficiency analysis, truth table construction, and derivation of the parsimonious solution—ensured methodological transparency.

Objectivity was sought by avoiding any researcher-generated scoring; instead, all conditions thresholds were derived by theoretical reasoning. The QCA model's pathways were interpreted cautiously given moderate raw coverage, the limited number of cases, and the high ratio of conditions to cases. Supplementary analysis using U5MR relative change provided robustness checks, yielding highly similar solution pathways.

Generalizability, while inherently limited in small size QCA, was upheld by the sample being all LIC/LMIC fragile states—cases with shared structural characteristics. Though not statistically generalizable, the findings identify plausible configurations relevant to comparable FCAS.

The integration of qualitative and quantitative approaches yielded a more comprehensive understanding of EDC in fragile settings than either method could alone. Qualitative findings revealed nuanced partnership dynamics shaped by political upheavals, humanitarian imperatives, and evolving system capacities. These insights go in line with the QCA results, helping explain why alignment and joint technical assistance emerged as sufficient but not necessary conditions for UHC progress in FCAS. This confirms that given the complexity of UHC, health systems and overall context within FCAS, change pathways are multiple. The QCA sub-study provided an empirical test of assumptions derived from Sudan's case,

demonstrating that improvements in cooperation behaviours can contribute to health outcomes. The complementary strengths of in-depth contextual analysis and cross-country configurational comparison offer a robust methodological foundation for examining EDC and UHC in FCAS.

Together, the four sub-studies advance a rigorous, trustworthy, and context-sensitive methodological approach to studying how EDC principles influence health systems towards their goals in fragile settings.

Positionality and reflexivity are crucial for this study uptake. Throughout the sub-studies papers (published or manuscripts) and this thesis, I tried to reflect openly on my position and background so the readers could be aware of any role, assumptions or judgement that entails. My role within Sudan health sector partnerships scene contributed to how this study is shaped. The existing good personal connections with various actors facilitated their high engagement rate. Yet my involvement at personal level could have hindered some responses. I tried to mitigate for such effect through follow up questions to ensure clarity of responses. Further, I must admit to my own personal stand on the importance of adhering to the EDC commitments as this was part of my professional mandate (see prologue). Stands of resistance to adhere or presenting challenges from partners (e.g. humanitarians) were negatively perceived by me, I believe, and this personal stand was weighed and tested to ensure objectivity through deliberations with the outsiders (Daniel and the team of supervisors). Lastly, being politically involved within Sudan different contexts presented, with strong leaning towards the 2019 revolution and transition forces, is important to report on for the readers' consideration.

Limitations

Due to the challenging changing context in Sudan, some aspects that we initially intended to cover within the study were not possible to address. The issue of mutual accountability was not feasible to investigate after the 2021 coup given the abrupt ending of communication between partners. Furthermore, after the first study, it became apparent that the CSOs had a limited active role in the partnership scene, and hence no further analysis on CSOs perspectives and role was carried out, unlike originally planned. The private sector was also weakly involved in the health sector deliberations including EDC agreements and the coordination platforms in Sudan, hence were not considered in the studies.

The lack of updated data for UHC reporting (only up to 2021) limited the ability to monitor progress over time. Further, the rating technique used for EDC conditions via personnel who have close relations with the SDG3 GAP partners, and the consideration of acquiescence bias limited the trust in the rating and lead to the choice of relying on the top rating only as the presence of the conditions. Moreover, due to the limited number of cases, I was not able to examine any further factors that could have affected the attainment of UHC. The limited number of cases, their similarity and the multiplicity of the conditions beyond the usually recommended threshold for QCA of around one condition per four cases (Mello, 2021) could have contributed to the low to moderate coverage of the final solutions, i.e. moderate level strength of explanation of the outcome at best. With further cases we could have not only improved this coverage issue but might have led to more deliberate solutions on UHC attainment pathways. Further, unfortunately the available report on the EDC conditions from the SDG3 GAP did not include date from Sudan. Hence, Sudan was not among the examined cases within sub-study 4.

The study preceded the Sudan war issued in April 2023. It was not feasible to collect further data within the scope and timeline of this Doctoral study. It was obvious at the beginning of the war that many of the partners were not clear on how they will be dealing with this situation. Yet, learning from the comparable periods, still less intensely negative, of Al-Bashir regime and the post-coup, could inform how things evolve in the war era. Empirical studies on the partnership scene during conflict periods is more available in the academic literature compared to what is presented within otherwise politically challenging Sudan partnership eras prior to 2023.

Conclusions and recommendations

Conclusions

This thesis explored how principles of EDC influenced health system governance and progress toward UHC in FCAS, using Sudan as a case study and complementing it with a cross-country comparative analysis. Sudan's experience illustrates that EDC adherence is highly contingent on political dynamics. Periods of openness, such as the post-2019 transition, facilitated alignment, revitalized coordination platforms, and encouraged system utilization. Conversely, political shocks, notably the 2021 coup, reversed these gains, freezing donor funds and dismantling joint planning processes. The findings showed that several EDC principles were individually sufficient to achieve health SDG targets. Partners alignment with national priorities and jointly planned technical assistance emerged as the most pivotal, enabling progress on strengthening health systems and achieving health targets in volatile FCAS context even when other conditions such as coordination or monitoring were imperfect. This reflects that despite the complexity of FCAS, multiple pathways can catalyse systemic improvements. The study affirms that technical interventions can make a difference, but alone they cannot overcome governance challenges as political legitimacy and openness remain critical anchors for cooperation.

Coordination platforms such as Sudan's Health Sector Partners Forum served as conversion spaces where principles translated into practice. When sustained through active investment from partners, these platforms fostered inclusive dialogue, joint planning, and piloting of system-use approaches. Their fragility under political stress highlights the need for resilient, institutionalized coordination structures that are government led. Persistent division between humanitarian and development actors was evident, particularly in attitudes toward system utilization. Humanitarian agencies prioritized speed and autonomy, often maintaining parallel systems, while development partners tipped more on the "use to strengthen" approaches towards the national health system. Technical assistance that is jointly planned among partners was a bridge that enabled phased integration of financial management, procurement, and information systems, and reducing perceived risks of system reliance. Such approach should be further emphasized within EDC agreements and progress evaluations.

Finally, the study identified significant gaps in monitoring and accountability platforms. Current global health partnership frameworks inadequately capture behavioural change in EDC implementation, relying mostly on perception-based indicators with infrequent assessments. Strengthening accountability through harmonized, practice-oriented metrics is essential for sustaining gains.

Overall, this thesis concludes that EDC offers a viable approach for strengthening health systems in FCAS, but its success depends on adaptive strategies that combine technical rigor with political sensitivity. Sustained progress requires institutionalized coordination, flexible technical assistance, and robust and aligned accountability mechanisms, all anchored in inclusive governance. While EDC cannot eliminate fragility's structural challenges, it can create reinforcing pathways that transform fragmented aid into resilient systems support, provided that global and local actors commit to principled, context-aware collaboration. In light of the current global changes in aid landscape with political pressure at donor countries for less aid spending, it is crucial to advocate for more effective approaches and advance the respect for alignment with real recipient countries priorities via joint and coordinated processes. One important element to ensure addressing real population priorities and embed resilience and sustainability assurances even during periods of political turmoil and legitimacy contestations is to empower the CSOs. Enhancing national CSOs capacities and engaging them in all partnership process entails prohibiting undermining the civil space by governments and global partners interventions. It is imperative to stress on aligning global partnership platforms and closely monitoring adherence to and progress on EDC principles as it is now more important than ever.

While the study outlines the importance of political dynamics, it does not promote a belief that actors working towards development and change should adhere to what is believed to be the boundaries set by the context. Rather the interventions need to be rooted in deep understanding of the political dynamics and try to act on bringing the various actors closer. Alignment with national health strategies in the complex FCAS setting requires risk-based safeguards that manage political, fiduciary, and operational risks while preserving national ownership. These safeguards encompass inclusivity of national planning (civil society centred), politically-sensitive analysis, adaptive implementation arrangements, gradual use of country systems, and joint accountability mechanisms.

Recommendations to improve EDC policy and practice

The following recommendations are addressed to global partners and recipient countries governments in FCAS (responsibility clarified in each).

1. **Clarify and institutionalize national ownership:**
    - Global partnership platforms and key entities (GPEDC, SDG3 GAP, UHC2030, WHO) must clarify the meaning of ownership in FCAS to avoid misuse specially within contested sovereignty situations.
    - Embed ownership in inclusive governance processes where governments involve and empower civil society, subnational authorities, and communities in planning and oversight.
2. **Enhance alignment and joint technical assistance:**
    - Global partners should align with nationally agreed health strategies and plans even in politically complex settings, using risk-based safeguards. This relies on inclusively developed plans.
    - Partners should develop and implement flexible, context-sensitive technical assistance that builds long-term capacity, and sustain the endeavours to reduce fragmentation and parallel systems.
3. **Sustain coordination mechanisms functionality:**
    - Government and partners must invest in resilient, inclusive platforms (like Sudan HSPF) with sustainable arrangements. Functionality under all circumstances even during political contestations allows having an anchor to build on afterwards.
    - Bridge the humanitarian–development divide and integrate humanitarian actors into all main coordination structures. Promote phased approaches to system utilization (by development partners and national governments) with risk mitigation measures (e.g., spot checks, phased integration) to enable humanitarian actors to engage with national system, building trust.
4. **Adapt to political realities:**
    - Partners must design EDC interventions with contingency plans for political shocks, via flexible fund and implementation modalities.
    - Partners and governments should empower civil society and subnational actors to enable sustainability of essential services and support even during shocks and periods of contested legitimacy.

Recommendations to enhance EDC monitoring and accountability

The following recommendations are directed to the global partnership platforms for health and beyond to the GPEDC.

1. Shift from perception-based metrics to EDC behavior change indicators that capture actual partner practices (e.g., proportion of funds channelled through national systems, frequency of joint TA planning, participation in inclusive coordination forums percentage).
2. Increase the frequency and autonomy of assessments by using digital dashboards and independent verification to reduce reporting bias.
3. Global health partnership frameworks must be harmonized to create a coherent accountability spine for EDC in health, ensuring alignment across UHC2030, SDG3 GAP, and related initiatives. The process should focus on learning from successes of IHP+ and revival of its comprehensive approach to EDC.

Recommendations for future research

1. Follow up examination of the mechanisms underlying the pathways from EDC to health system strengthening and development targets. Methods like realist evaluation could enable this examination.
2. Analysis focused on progress on UHC due to advancements in EDC implementation (as new data comes on the EDC conditions and UHC).
3. Investigate uncovered EDC aspects like accountability in FCAS.
4. Explore further the partnership and roles played by the civil society organizations in FCAS and how they adapt and interact with others within complex and volatile systems.

Acknowledgements

Reaching the end of this doctoral journey has been possible only because of the remarkable people who supported me, guided me, and believed in me through every challenge and turning point. Crossing path with outstanding people was the best blessing I enjoyed throughout my life.

To my supervisors and co-authors (Miguel, Anni-Maria, and Anna-Karin) and my examiner Isabel Goicolea, thank you for your unwavering guidance, backing, patience, trust and intellectual generosity. Your mentorship has shaped not only this thesis but also the scholar I am becoming. I remain deeply grateful for your support, encouragement, and for pushing me to grow beyond my own expectations.

My family has been my greatest source of strength through this journey. I am forever grateful to **my parents (Jabir and Fatima)**, whose prayers, sacrifices, and belief in education built the foundation upon which every achievement rests. Thank you for giving me everything. To **Lina and Jude**, your support, understanding and resilience, sustained me through the stress, the absences, long nights, and all difficult phases. This accomplishment is as much yours as it is mine. To my broader family—siblings, uncles, and aunts—your constant support, messages, and moral encouragement kept me grounded and connected to home.

My profound gratitude goes to the *department of epidemiology and global health at Umeå University* that welcomed me and supported me throughout this journey. Its special leadership and exceptional environment made it possible to navigate this process despite difficulties. I especially thank the administrators' group, with a distinct mention for **Ulrika Harju.** You make it all work Ulrika. Thanks for everything. Without any doubt I thank the whole department staff, but I can't forget **Karin Johansson and Ulrika Järvholm** for their kindness and care. You might have left the department before my big day, but your support has left traces that time cannot erase.

A special thank you to **Dr. Imad Eldin Kayona** who has been my mentor and supervisor though my journey at Sudan Ministry of Health and WHO-Sudan and extended to a part of this doctoral journey. The gratitude is extended to the *WHO-Sudan office*, with special mention to **Dr. Naeema al-Gasseer and Dr Nima Saeed Abid.** I am grateful for enabling and supporting my doctoral studies under your wise leadership.

Duly, I am grateful to WHO seniors and colleagues who encouraged and inspired me. Thanks **Dr. Kamran Mashhadi, Dr. Mohamed Yusuf, Dr. Muna Ismail, Dr. Salim Mohamednour, Dr. Faihaa Ahmed, Dr. Ahmed Esawi** and all the journey colleagues for your belief in my potential, your advice, guidance, and all the experiences we lived through.

To my ***EpiGH friends***, the ones who stayed close across borders and time zones, and those who became family along the way, you made this journey meaningful, and joyful. **Mazen, Camila, Cartrine, Nuria, Isabelle Hempler, Maria, Daniel, Kaspar, Hanna, Fethi, Angela, Elia, Ester, Alieu, Kapuya, Pebalo, Andre, Albert, Carla, Chaibo, Adam, Chama, Joseph, Kirabo, Nahum, Paul**, **Puthy, Rebecca, Rodrigo, Yosephine,** and all. thank you for your companionship, insights, and the countless moments of academic and personal dialogue. Listing everyone at EpiGH, past and present, would have been optimum. I did not mean to omit anyone. The listed were colleagues at several stages.

My heartfelt thanks to my ***Umeå Sudanese family***, especially **Heba, Modber, Amin, Abdalla, Hussein, Ramiz, Mohamed Adam,** and the rest all of whom touched my life here with care, warmth and solidarity. Your presence in my life during this period meant more than you know.

Special greetings to friends who supported in academic ways within this thesis. Thanks, **Daniel Tsang**, for your sincere support and contribution, and **Amel Elfatih** for a friendship that extended far beyond shared work. You inspired my sub-study on the nexus and initiated it with me, while being the best work mate.

My sincere appreciation goes to my *study respondents* for their valuable contributions. I am also grateful to the *reviewers* at each milestone of my doctoral study -admission, mid-term, and pre-defence- for their feedback, critical insights, and for helping refine and strengthen my research work.

This study would not have been possible without the generous support of *Erling-Persson Foundation* that awarded me the scholarship for 'Forming New Leaders in Global Health'. I hope I rise to that aspiration.

Finally, I am grateful to everyone who supported me in any way through conversations, prayers, kindness, or simply believing that I would reach this goal. I am truly sorry for not mentioning any special name. This thesis is a testament not only to academic effort but to the power of care, compassion, and perseverance.

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Appendices

Annex 1. Interview guide: Sudan health sector partners relations and adherence to EDC principles

Interview No: …

Introduction

There is an international consensus on effective development cooperation principles that can assist in reaching development targets, including those for healthy lives and wellbeing. Sudan is not an exception to these agreements, and a local health compact is in place as well as corresponding strategy. Proper attention to co-operation principles of alignment and harmonization is central in all EDC agreements. The aim of this interview is to get an insight on your views as a health actor in Sudan on the partners relations, perspectives on these concepts and the implementation of these EDC commitments at the country level (Sudan).

Checklist

Interview duration: … minutes.
Interview place: ……
Request to record the interview: Approved or not
Explanation of how information will be handled with confidentiality: Explained or not
Request for permission to quote: Approved or not
Name of interviewee: **……**
Interviewee agency: **……**
Function: **……**
Date and time of interview: **……**
Written consent: Signed or not

❖ Questions in red are not to be asked for Government respondents.

❖ Grey write-up is follow-up questions in case not covered in response.

Opening questions: Strategic priorities and ownership

In which capacity and fields do you deal with health sector development cooperation in Sudan? Since when? What have your fields of work been during the last five years? And has this witnessed any changes recently?

What are in your view the development priorities of Sudan health sector? have these changed from the priorities 5 years ago?

From your experience in Sudan, how are Sudan health strategies developed? How are priorities identified and who is involved in this process? Have there been any changes in this domain during the last 6 years?

Who is currently leading and mainly influencing the process of strategic priorities identification and how? In your opinion, who should lead the process of priorities identification in Sudan health sector? and why? Has this been in the situation for the last 5 years or has this witnessed some changes? Please elaborate if yes

How is ownership exercised and viewed by the various stakeholders?

Alignment and harmonization

Can you describe the process in use now for the health partners to align their development cooperation? (Do the partners regularly meet to discuss programs alignment issues? Is there a partner leading the process? Which partner? How? What is your view on this process and how to improve it? Mention any changes seen)

What is the set up / platform in use to co-ordinate between health sector development partners in Sudan? (Are there any agreements on cooperation between partners? Do you know if partners regularly meet to discuss their cooperation and joint work? Are you aware and involved in the process of these meetings? How often did you meet with other health sector partners in Sudan in workshops, conferences or meetings? What was the scope of the issues discussed? What are the facilitation arrangements? Who leads and who is active in facilitating this process? What can be improved and how? Has there been any progress on this?)

What instruments did you bring to align your work with the national strategies and priorities? (Do you have tools that enable you to align your work behind the identified national needs and priorities? Since when?)

What role does the government of Sudan currently play to enhance partners alignment of their work with the national strategic priorities? Has it always played the same role, or has this changed from 5 years ago? (Does the government of Sudan FMOH has a complete and up to date document through which it sets out its health strategy and priorities?

What are the priorities/needs identified in this document? Are there clear targets that enable partners to align their programming on the needs and priorities identified by the government?)

What challenges do you face in aligning your work with the priorities of Sudan health sector? please mention if these challenges changed during last 6 years and if you see any opportunities for improvement.

Are you now able to use the country systems in delivering your support? (If not ask why. Has this changed?)

What efforts are going to ensure strengthening and utilizing the country's systems in the cooperation if any? (Refer to specific systems of financial management, procurement and information and take 5 years span.)

Have the health sector partners agreed on a process for harmonized systems for funding modalities, financial management and procurement? If yes, describe it please and the progress on it so far.

Do you share all information on your work with the other sector partners? (If not ask why). In which ways?

Have there been regular joint reviews during the last five years? Elaborate on how it was done or why not.

Have any actions been taken to reduce the number of separate reviews for the priority sector interventions?

Has your agency formulated/updated an official policy on harmonization between partners? If yes, describe it please and explain who is/are responsible for leading the implementation of this harmonization policy?

What are the challenges you face in utilizing the country systems of PFM, procurement and information? Have they changed during last 6 years? Kindly advise on any opportunities you see in this regard.

Are there other factors influencing the implementation of these agreed-upon commitments to align and harmonize the work in the health sector as well as use national systems discussed earlier? Please elaborate

Partnership and relations

How do you perceive the relations between the health partners in Sudan? Are any partners dominating or being marginalized? How and why? What drives this influence? Has this changed over the last 5 years?

How have the various health partners reacted to each other's interests and power during the past five years? Do you think this has / will affect the implementation of EDC practices and how? How can it be improved?

What is your view on the future prospects of EDC in Sudan health sector? how do you believe the changing political context is influencing EDC roll-over? Are EDC commitments having more or less attention? Why?

Concluding remarks

How do you view the Global Action Plan within this context? What are the challenges you perceive the GAP will face and what are the opportunities it can bring and can assist its implementation success?

Do you have any lessons from the past for GAP and future EDC interventions to reflect on? Please elaborate

Is there anything you want to add to this interview?
Ask if the interviewee knows others who are important to interview. Why and if interviewee can facilitate the process?
Ask for any relevant documents
Thank the respondent.

Annex 2. Interview guide: Humanitarian–development nexus approach to health systems strengthening in Sudan

Interview No: ...

Introduction

Thank you for agreeing to join this study. The study aims to explore the understanding and conceptualization process of the humanitarian-development nexus approach to health system strengthening in Sudan.

Checklist

Interview duration: ... minutes.
Interview place: .....
Request to record the interview: Approved or not
Explanation of how information will be handled with confidentiality: Explained or not
Request for permission to quote: Approved or not
Name of interviewee: .....
Interviewee agency: .....
Function: .....
Date and time of interview: .....
Written consent: Signed or not

Nexus Understanding, introduction and conceptualization

How do you perceive the concept of the nexus?

How did you get familiar with the nexus concept?

How was it introduced to the health sector? When and by who?

Can you describe your view on the status of partnership for health for both humanitarian and development partners?

How do you see the progress in the nexus use in Sudan health sector?

The nexus in practice in Sudan health sector

How does your organization/entity perceive and act on the nexus concept?

Has this been affected by any contextual factors? Kindly elaborate

Are there any players who lead and coordinate the operationalization of the nexus in Sudan? Has this changed overtime?

What are the changes or paradigm shifts necessary to enable the nexus approach? What can be the enablers and motives for these paradigm shifts?

Has this been affected by context changes?

Have your organization/entity embarked on adopting any nexus linked project(s)? kindly describe and elaborate any changes with context shifts in Sudan.

Are you aware of any practices and projects that are relevant to the nexus concept carried out by other partners? If yes, describe those projects and all known progress

Were there any platforms that influenced your conceptualization and action on the nexus? Which and how?

The nexus and health system strengthening and utilization

What efforts are going to ensure strengthening and utilizing the country's systems in the cooperation if any? (Refer to specific systems of financial management, procurement and information and take 5 years span.)

Do you believe health partners are able to use the country systems in delivering their support? (If not ask why. Has this changed?)

Have the health sector partners agreed on a process for harmonized systems for funding modalities, financial management and procurement? If yes, describe it please and the progress on it so far.

Have there been regular joint reviews during the last five years? Elaborate on how it was done or why not.

Have any actions been taken to reduce the number of separate reviews for the priority sector interventions?

Do you share all information on your work with the other sector partners? (If not ask why). In which ways?

What are the challenges you face in utilizing the country systems of PFM, procurement and information? Have they changed during last 6 years? Kindly advise on any opportunities you see in this regard.

Challenges and opportunities for the nexus implementation in Sudan health sector

What are the main challenges you experience related to the implementation of the nexus and nexus inspired projects? (Explore the root causes behind those challenges)

What can be done to mitigate those challenges and the contextual factors?

What opportunities do you foresee to support the nexus implementation within Sudan health sector?

Concluding remarks

Is there anything you want to add to this interview?

Ask if the interviewee knows others who are important to interview. Why and if interviewee can facilitate the process?

Ask for any relevant documents

Thank the respondent

Annex 3. Comparative analysis for sub-study 4: full data set and extra analysis details

Table S1.1: Full data set (uncalibrated)

Country	align	Partners coordinate	National coordination	budget	National M&E	Joint TA	UHC
1. Afghanistan	3	3	3	2	3	2	40.9
2. Burundi	4	4	3	4	3	3	41.5
3. Chad	4	3	2	3	4	4	29.4
4. Comoros	5	4	4	4	3	3	48.1
5. DRC	5	5	4	5	4	3	41.7
6. Eritrea	5	5	5	5	5	3	44.9
7. Ethiopia	4	4	5	4	5	4	35.1
8. Haiti	3	2	2	2	2	2	54.1
9. Mali	4	4	5	4	5	5	41.3
10. Mozambique	4	3	3	3	2	3	44.0
11. Niger	4	4	4	4	3	4	35.0
12. Nigeria	4	4	4	4	4	4	38.4
13. Somalia	3	2	1	2	2	2	26.8
14. South Sudan	3	3	4	4	4	2	34.2
15. Syrian Arab Republic	4	4	3	3	4	4	64.1
16. Yemen	3	3	1	2	1	4	42.5
17. Zimbabwe	4	4	3	3	3	5	55.0

Table S1.2: Sufficiency and Necessity Analysis of negative outcome and negative conditions

Condition	Necessity analysis		Sufficiency analysis	
	Consistency	***Coverage***	***Consistency***	***Coverage***
~Align	1.000	0.429	0.429	1.000
~Partners coordinate	1.000	0.400	0.400	1.000
~National coordination	0.833	0.357	0.357	0.833
~Budget	1.000	0.400	0.400	1.000
~National M&E	0.833	0.357	0.357	0.833
~Joint TA	1.000	0.400	0.400	1.000

Interpretation:

The negative conditions on alignment, partners coordinate, budget and Joint TA have consistency of 1 with ~UHC. Their necessity coverage is low.

No single condition in the negated form is sufficient for the attainment of the negated UHC.

Table S1.3: Truth table

```
##   OUT: output value
##     n: number of cases in configuration
##  incl: sufficiency inclusion score
##   PRI: proportional reduction in inconsistency
```

	alignment.cr	coordinate.cr	coordinationmech.cr	budget.cr	nationalme.cr	jointa.cr	OUT	n	incl	PRI	cases
3	0	0	0	0	0	1	1	1	1.000	1.000	17
8	0	0	1	0	1	1	1	1	1.000	1.000	9
33	1	0	0	0	0	0	1	1	1.000	1.000	4
57	1	1	0	1	0	0	1	1	1.000	1.000	5
62	1	1	1	1	1	0	1	1	1.000	1.000	6
1	0	0	0	0	0	0	0	11	0.545	0.545	1,2,3,8,10,11,12,13,14,15,16
6	0	0	1	0	1	0	0	1	0.000	0.000	7

Interpretation of the truth table:

While there could be 64 possible configurations (2^6), our dataset had five configurations leading to the outcome, two linked to lack of the outcome, and the rest of the configurations were not present within our dataset.

Annex 4. Supplementary analysis for EDC principles toward improving Under Five Mortality

This analysis was done using under-five mortality rate relative change (U5MRC) 2016-2022, which was calculated based on the formula below:

$$\text{U5MRC} = \frac{U5MR2016 - U5MR2022}{\text{U5MR2016}} * 100$$

Table S2.1: Complete data set (uncalibrated)

Country	align	Partners coordinate	National coordination	budget	national M&E	joint TA	U5MRC
1. Afghanistan	3	3	3	2	3	2	17.81
2. Burundi	4	4	3	4	3	3	21.09
3. Chad	4	3	2	3	4	4	17.68
4. Comoros	5	4	4	4	3	3	18.44
5. DRC	5	5	4	5	4	3	18.18
6. Eritrea	5	5	5	5	5	3	18.49
7. Ethiopia	4	4	5	4	5	4	22.87
8. Haiti	3	2	2	2	2	2	16.67
9. Mali	4	4	5	4	5	5	16.25
10. Mozambique	4	3	3	3	2	3	16.62
11. Niger	4	4	4	4	3	4	6.53
12. Nigeria	4	4	4	4	4	4	13.90
13. Somalia	3	2	1	2	2	2	17.43

14. South Sudan	3	3	4	4	4	2	0.00
15. Syrian Arab Republic	4	4	3	3	4	4	38.33
16. Yemen	3	3	1	2	1	4	13.78
17. Zimbabwe	4	4	3	3	3	5	17.04

For the U5MRC threshold we calculated the mean (found to be 17.12), which we used then as the cut-off point.

Tables S2.2: Necessity and Sufficiency Analysis

S2.2.1: Necessity Analysis – positive outcome

```
QCAfit(data.cr[,2:7], data.cr$U5MRC.cr, necessity = TRUE, neg.out = FALSE)
```

```
##                          Cons.Nec Cov.Nec   RoN*
## Align.cr                    0.333   1.000 1.000
## partners coordinate.cr      0.222   1.000 1.000
## budget.cr                   0.222   1.000 1.000
## national M&E.cr             0.222   0.667 0.933
## joint TA.cr                 0.222   1.000 1.000
## national coordination.cr    0.222   0.667 0.933
## ~Align.cr                   0.667   0.429 0.273
## ~partners coordinate.cr     0.778   0.467 0.200
## ~budget.cr                  0.778   0.467 0.200
## ~national M&E.cr            0.778   0.500 0.300
## ~joint TA.cr                0.778   0.467 0.200
## ~nationalcoordination.cr    0.778   0.500 0.300
```

Comment: No necessary condition
*RoN: Relevance of Necessity (measure of necessity trivialness if <.5)

S2.2.2: Necessity Analysis – Negative outcome

```
QCAfit(data.cr[,2:7], data.cr$U5MRC.cr, necessity = TRUE, neg.out = TRUE)
```

```
##                           Cons.Nec Cov.Nec   RoN
## Align.cr                     0.000   0.000 0.824
## partners coordinate.cr       0.000   0.000 0.882
## budget.cr                    0.000   0.000 0.882
## national M&E.cr              0.125   0.333 0.875
## joint TA.cr                  0.000   0.000 0.882
## national coordination.cr     0.125   0.333 0.875
## ~Align.cr                    1.000   0.571 0.333
## ~partners coordinate.cr      1.000   0.533 0.222
## ~budget.cr                   1.000   0.533 0.222
## ~national M&E.cr             0.875   0.500 0.300
## ~joint TA.cr                 1.000   0.533 0.222
## ~national coordination.cr    0.875   0.500 0.300
```

Comment:

The negatives for Align, partners coordinate, budget and Joint TA are **NECESSARY** conditions to not achieve a good change in U5MR when all conditions are set at 5 (strongly agree) within this dataset. This covers 53-57% of the cases that didn't achieve change.

S2.2.3: Sufficiency Analysis – positive outcome

```
QCAfit(data.cr[,2:7], data.cr$U5MRC.cr, necessity = FALSE, neg.out = FALSE)
```

```
##                           Cons.Suf Cov.Suf  PRI**
## Align.cr                     1.000   0.333  1.000
## partners coordinate.cr       1.000   0.222  1.000
## budget.cr                    1.000   0.222  1.000
## national M&E.cr              0.667   0.222  0.667
## joint TA.cr                  1.000   0.222  1.000
## national coordination.cr     0.667   0.222  0.667
## ~Align.cr                    0.429   0.667  0.429
## ~partners coordinate.cr      0.467   0.778  0.467
## ~budget.cr                   0.467   0.778  0.467
## ~national M&E.cr             0.500   0.778  0.500
## ~joint TA.cr                 0.467   0.778  0.467
## ~national coordination.cr    0.500   0.778  0.500
```

Comment:

Align, partners coordinate, budget and Joint Technical Assistance are sufficient to achieve good U5MR change or at least can be part of an expression that is sufficient. 22-33% of the change can be explained by these conditions within the successful cases.

**PRI: Proportional Reduction in inconsistency (condition with PRI<.5 are not sufficient even if consistency threshold is met).

S2.2.4: Sufficiency Analysis – Negative outcome

```
QCAfit(data.cr[,2:7], data.cr$U5MRC.cr, necessity = FALSE, neg.out = TRUE)
```

```
##                           Cons.Suf Cov.Suf   PRI
## Align.cr                     0.000   0.000 0.000
## partners coordinate.cr       0.000   0.000 0.000
## budget.cr                    0.000   0.000 0.000
## national M&E.cr              0.333   0.125 0.333
## joint TA.cr                  0.000   0.000 0.000
## national coordination.cr     0.333   0.125 0.333
## ~Align.cr                    0.571   1.000 0.571
## ~partners coordinate.cr      0.533   1.000 0.533
## ~budget.cr                   0.533   1.000 0.533
## ~national M&E.cr             0.500   0.875 0.500
## ~joint TA.cr                 0.533   1.000 0.533
## ~national coordination.cr    0.500   0.875 0.500
```

Comment:
No sufficient condition

Table S2.3: Truth table

```
TT <- truthTable(data.cr, outcome = U5MRC.cr,  conditions = c("Align.cr", "partners coordinate.cr", "budget.cr",
"national M&E.cr", "joint TA.cr", "national coordination.cr"),         show.cases  = TRUE, sort.by = "incl, n")
TT
##   OUT: output value
##     n: number of cases in configuration
##  incl: sufficiency inclusion score
##   PRI: proportional reduction in inconsistency
##
##    Align.cr partners coordinate.cr budget.cr nationalM&E.cr joint TA.cr
##  3         0              0           0            0            1
##  8         0              0           0            1            1
## 33         1              0           0            0            0
## 57         1              1           1            0            0
## 62         1              1           1            1            0
##  1         0              0           0            0            0
##  6         0              0           0            1            0
##   national coordination.cr OUT    n   incl  PRI   cases
##  3            0             1     1   1.000 1.000 5
##  8            1             1     1   1.000 1.000 15
## 33            0             1     1   1.000 1.000 6
## 57            0             1     1   1.000 1.000 13
## 62            1             1     1   1.000 1.000 4
##  1            0             0     11  0.364 0.364 1,2,3,7,8,9,10,11,14,16,17
##  6            1             0     1   0.000 0.000 12
```

Table S2.4: Parsimonious Solution

```
##
## M1: Align.cr + joint TA.cr -> U5MRC.cr
##
##                   inclS   PRI   covS   covU   cases
## ----------------------------------------------------
## 1  Align.cr       1.000  1.000  0.333  0.333  Comoros, DRC, Eritrea
## 2     joint TA.cr 1.000  1.000  0.222  0.222  Mali, Zimbabwe
## ----------------------------------------------------
##              M1  1.000  1.000  0.556
```

Comment:

The coverage measures for sufficiency and final model pathway towards U5MR relative change were slightly higher than those for the UHC outcome yet both remain to have moderate coverage. The separately low coverage of each of the sufficient conditions increases to their coverage sum in the model, indicating that there are no cross-over areas between them (evident in truth tables).